

REVIEW

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Decoding breast cancer: insights into molecular pathways & therapeutic approaches

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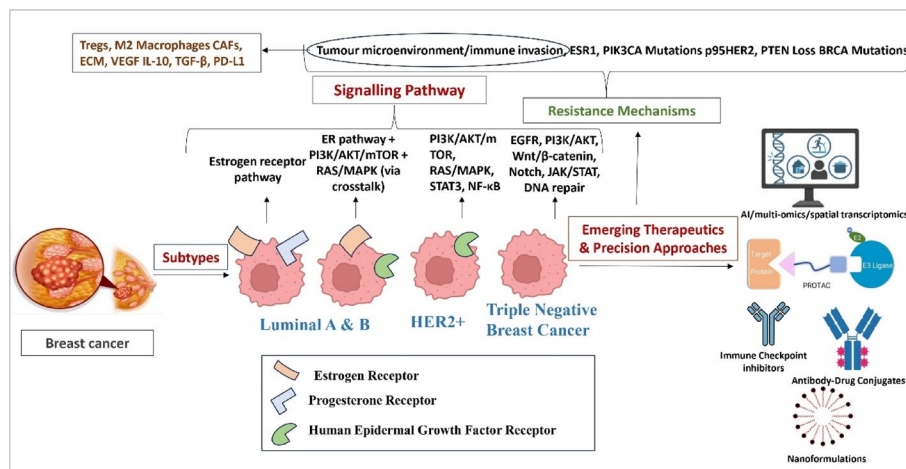
Abstract

Breast cancer is a highly heterogeneous malignancy encompassing distinct molecular subtypes such as Luminal A & B, HER2 enriched and triple negative breast cancer, each characterized by unique genomic, transcriptomic and proteomic landscapes that dictate differential prognosis and therapeutic response. This review delineates the complex signalling pathways underlying breast cancer including estrogen receptor, progesterone receptor and HER2 pathways and their downstream effectors such as PI3K/AKT/mTOR, RAS/MAPK, Notch, Wnt/ β -catenin and JAK/STAT cascades. The interconnectivity of these pathways not only governs tumor cell proliferation, survival and metastasis but also contributes to intrinsic and acquired resistance mechanisms against standard-of-care therapies including endocrine agents, HER2-targeted antibodies and chemotherapy. Emerging therapeutic strategies such as next-generation selective estrogen receptor degraders, proteolysis-targeting chimeras, antibody-drug conjugates, immune checkpoint inhibitors and pathway-specific small-molecule inhibitors highlights their mechanistic rationale, clinical evidence and biomarker-guided applicability. Resistance mechanisms involving ESR1 mutations, PTEN loss, p95HER2 expression, BRCA reversions and tumor microenvironment-mediated immunosuppression contributes to therapy resistance. The review explores molecular subtypes, oncogenic signalling pathways, treatment approaches, resistance mechanisms including acquired resistance & crosstalk, metabolic reprogramming, epigenetic modifications, immune modulations and tumor microenvironment, spatial transcriptomics insights into immune cell subsets & treatment response biomarkers and the role of artificial intelligence in breast cancer treatment. The integration of artificial intelligence-based predictive modelling and rational drug combinations offers a promising paradigm for overcoming resistance and tailoring precision oncology approaches. By comprehensively mapping these oncogenic factors and therapeutic intersections, this review provides a strategic framework for optimizing breast cancer management in the era of molecular medicine.



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Graphical abstract



Keywords Crosstalk, Acquired resistance, Therapeutic implications, Spatial transcriptomics, AI

1 Introduction

Breast cancer remains one of the most prevalent malignancies globally, affecting millions of women and a smaller proportion of men annually, with its incidence driven by diverse genetic, molecular and environmental factors [1]. Its heterogeneity, characterized by distinct molecular subtypes and signalling pathways poses significant challenges to effective diagnosis and treatment [2]. Understanding the molecular underpinnings of BC is critical for developing targeted therapies and overcoming resistance, a major obstacle in clinical management [3]. BC is categorized into molecular subtypes such as Luminal A, Luminal B, HER2-enriched and TNBC, based on the expression of ER, PR and HER2. These subtypes exhibit unique biological behaviors and clinical outcomes, necessitating tailored therapeutic approaches including endocrine therapies, HER2-targeted agents and chemotherapy [4]. Luminal A and B tumors which are hormone receptor-positive (ER + and/or PR+), account for 70–80% of cases and rely heavily on ER signaling for growth [1]. Luminal A tumors are characterized by low proliferation and high hormone dependency, responding well to endocrine therapies like SERMs and aromatase inhibitors [5]. In contrast, Luminal B tumors display higher proliferation rates and frequent co-expression of HER2 or EGFR, leading to a more aggressive phenotype and partial resistance to endocrine therapy due to pathway crosstalk and ESR1 mutations [6, 7]. HER2-enriched tumors, comprising 15–20% of BC cases are driven by ERBB2 gene amplification, resulting in aggressive proliferation via the PI3K/AKT/mTOR and RAS/MAPK pathways [1]. Therapies such as trastuzumab and ADCs like T-DXd have revolutionized outcomes, though resistance, often mediated by truncated HER2 forms (p95HER2) or compensatory pathways remains a challenge [8, 9]. TNBC, representing 10–15% of cases, lacks ER, PR and HER2 expression, making it highly heterogeneous and reliant on chemotherapy due to limited targeted options [1]. Recent advances, however, have identified actionable targets like BRCA1/2 mutations and immune checkpoint pathways, expanding therapeutic possibilities [10, 11]. The molecular pathogenesis of BC

is orchestrated by key oncogenic signaling pathways including ER, HER2, PI3K/AKT/mTOR, RAS/MAPK, Wnt/ β -catenin, JAK/STAT and Notch [2]. ER signaling drives proliferation in ER + tumors through genomic and non-genomic mechanisms, with resistance arising from ESR1 mutations and alternative pathway activation [5, 6]. HER2 signaling, activated by receptor dimerization promotes aggressive tumor behavior, with resistance linked to PI3K/AKT/mTOR activation and p95HER2 [8]. The PI3K/AKT/mTOR pathway, dysregulated in HR + and HER2 + tumors, supports tumor survival and is targeted by inhibitors like alpelisib [7, 12]. The RAS/MAPK pathway, critical in HER2 + and TNBC drives proliferation and is inhibited by MEK inhibitors [13]. Wnt/ β -catenin and Notch pathways contribute to CSC maintenance and EMT particularly in TNBC, while JAK/STAT signaling promotes immune evasion [14, 15]. Crosstalk among these pathways amplifies resistance, necessitating combination therapies like dual PI3K/MEK inhibition [13, 16]. Emerging technologies such as spatial transcriptomics and artificial intelligence are reshaping BC research and treatment [17, 18]. Spatial transcriptomics reveals TME dynamics, identifying immune cell interactions and heterogeneity to guide immunotherapy strategies [17]. AI-driven approaches enhance biomarker discovery, predict therapeutic resistance and optimize personalized treatment by integrating multi-omics data [18, 19]. Additionally, nanoformulations and synergistic therapies including kinase inhibitors combined with ICIs, improve drug delivery and efficacy particularly in resistant subtypes like TNBC [20, 21]. This review comprehensively examines the molecular mechanisms underpinning BC subtypes, their therapeutic implications and strategies to overcome resistance, emphasizing precision medicine. By integrating insights from clinical trials and cutting-edge technologies, it highlights current advancements in BC management that address the disease's molecular complexity [4, 10, 19].

2 Molecular subtypes

Advances in genomic profiling have revealed distinct molecular subtypes such as luminal A, luminal B, HER2-enriched, basal-like, and normal-like, each with unique gene expression patterns and therapeutic implications [22, 23]. BC is categorized into molecular subtypes such as Luminal A, Luminal B, HER2-enriched, and TNBC, based on the expression of ER, PR, and HER2. These subtypes guide clinical decisions, particularly in determining hormone receptor status for targeted therapies [24]. These subtypes exhibit unique biological behaviors and clinical outcomes, necessitating tailored therapeutic approaches, including endocrine therapies, HER2-targeted agents, and chemotherapy [25]. Understanding the molecular underpinnings of BC is critical for developing targeted therapies and overcoming resistance, a major obstacle in clinical management [1]. A brief detail of each subtype is discussed below:

2.1 Luminal A & B

Luminal A and B tumors are characterized by the expression of hormone receptors, specifically ER and/or PR which drive their biology [25]. Luminal A tumors typically exhibit high ER and PR expression, low proliferation rates (indicated by low Ki-67 levels) and a gene expression profile resembling normal luminal epithelial cells of the breast. They often express genes such as ESR1 (encoding ER α), GATA3 and FOXA1 which regulate epithelial differentiation and hormone responsiveness. Luminal B tumors, while also ER-positive, are distinguished by higher proliferation rates (elevated Ki-67), lower PR

expression and frequent co-expression of growth factor receptors like HER2 or (EGFR [1, 22]. This subtype often shows amplification of proliferation-associated genes such as CCND1 (cyclin D1) and MYC, contributing to a more aggressive phenotype compared to Luminal A [1]. The molecular pathogenesis of luminal tumors is driven by ER signaling where estrogen binds to ER, leading to dimerization, nuclear translocation and binding to estrogen response elements on DNA. This activates transcription of genes involved in cell proliferation (CCND1), survival (BCL2) and differentiation [22]. In Luminal B tumors, additional pathways like PI3K/AKT/mTOR and RAS/MAPK are frequently activated due to crosstalk with growth factor receptors, enhancing proliferation and contributing to therapeutic resistance [26, 27]. Luminal A tumors generally respond well to endocrine therapy due to their low proliferative index and strong hormone dependency, with 5-year survival rates exceeding 90% [22]. Luminal B tumors, however, exhibit poorer responses due to higher proliferation and pathway crosstalk. For example, ESR1 mutations (e.g., Y537S, D538G) in Luminal B tumors can lead to ligand-independent ER activation, reducing the efficacy of SERMs and aromatase inhibitors [27]. Additionally, upregulation of PI3K/AKT/mTOR signaling, often due to PIK3CA mutations (present in ~ 40% of ER + tumors) contributes to resistance by providing alternative survival signals [24].

2.2 HER2-enriched tumors

HER2-enriched tumors, accounting for 15–20% of BC cases are defined by amplification or overexpression of the ERBB2 gene encoding the HER2, a tyrosine kinase receptor in the EGFR family [23, 28]. HER2 amplification leads to ligand-independent receptor dimerization, activating downstream pathways like PI3K/AKT/mTOR and RAS/MAPK which drive aggressive tumour proliferation, survival and invasion [26]. These tumors often lack significant ER/PR expression, distinguishing them from Luminal subtypes, though some overlap exists with Luminal B tumors co-expressing HER2 [28]. Gene expression profiling reveals upregulation of proliferation genes (CCND1, MYC) and downregulation of apoptosis-related genes (BCL2), contributing to their high proliferative index and poor prognosis without treatment [22, 23]. HER2 signaling is initiated by receptor homo- or heterodimerization (e.g., with EGFR or HER3), leading to autophosphorylation and activation of pathways promoting cell cycle progression (CCND1), anti-apoptosis (BCL2) and angiogenesis (VEGF) [1]. Mutations in PIK3CA (~ 30% of HER2 + tumors) and loss of PTEN further enhance PI3K/AKT signaling, exacerbating tumor aggression and resistance [24, 26].

2.3 Triple-negative breast cancer

TNBC, comprising 10–15% of BC cases is defined by the absence of ER, PR and HER2 expression, making it a diagnosis of exclusion [24]. TNBC is highly heterogeneous, with subtypes like basal-like (70–80%, expressing basal cytokeratins and EGFR), mesenchymal and luminal androgen receptor [22, 28]. Its aggressive behavior stems from high proliferative rates, genomic instability (tumor protein 53 mutations in 80%) and frequent BRCA1/2 mutations (20%) [22, 26]. TNBC exhibits a basal-like gene expression profile with upregulation of proliferation (MKI67), EMT (SNAIL, SLUG) and angiogenesis (VEGF) genes, contributing to early metastasis and poor prognosis [22, 28]. The lack of hormone receptors and HER2 overexpression limits targeted therapy options,

historically relying on chemotherapy [22]. Molecular drivers include the PI3K/AKT/mTOR pathway which is activated in ~ 25% of TNBCs due to PIK3CA mutations or PTEN loss, promoting survival and resistance [25]. DNA repair defects due to BRCA1/2 mutations impair homologous recombination repair, increasing genomic instability but conferring sensitivity to DNA-damaging agents and immune infiltration as TNBCs often show high TILs, particularly in basal-like subtypes, suggesting potential immunotherapy responsiveness [25, 26, 28]. The absence of ER, PR and HER2 expression eliminates the efficacy of endocrine therapies and HER2-targeted agents, making cytotoxic chemotherapy (e.g., anthracyclines, taxanes) the standard of care [22]. However, TNBC's high recurrence rate (30–40% within 5 years) and metastatic potential necessitate novel approaches [3]. The lack of defined molecular targets has historically limited precision medicine but recent advances have identified actionable vulnerabilities [26, 28].

3 Key oncogenic signalling pathways

3.1 Estrogen receptor signaling pathway

ER signaling is a pivotal driver in the development and progression of ER + BC, accounting for approximately 70–80% of all cases [29]. The process begins when 17 β -estradiol, a primary estrogen diffuses across the cell membrane and binds to ER α (encoded by ESR1), a nuclear receptor predominantly located in the cytoplasm of BC cells, with high expression in Luminal A tumors and moderate expression in Luminal B tumors [30]. This binding induces a conformational change in ER α , releasing chaperone proteins such as heat shock protein 90 and activating the receptor [31]. Activated ER α monomers then dimerize, forming homodimers (ER α /ER α) or less commonly heterodimers with ER β , a step essential for DNA interaction and transcriptional activity [32]. In Luminal B tumors, co-expression of growth factor receptors (e.g., EGFR, HER2) enhances ER activity through crosstalk, amplifying signaling. The estrogen-ER complex translocates to the nucleus, facilitated by nuclear localization signals within ER, enabling it to function as a transcription factor. In the nucleus, the ER dimer binds to specific DNA sequences called estrogen response elements in the promoter regions of target genes, recruiting co-regulatory proteins such as co-activators (e.g., AIB1 or SRC family members) or co-repressors to modulate transcriptional activity [31, 33]. This interaction drives the expression of genes critical for BC progression, including CCND1 (encoding cyclin D1) for cell cycle progression, BCL2 for anti-apoptosis and MYC for proliferation, with Luminal A tumors showing strong ER dependence and Luminal B tumors exhibiting additional MYC activation due to higher Ki-67 levels [29–31]. In addition to this classical genomic pathway, non-genomic ER signaling occurs via membrane-bound ERs or G protein-coupled estrogen receptor, rapidly activating pathways like PI3K/AKT and MAPK/ERK which amplify proliferative and survival signals [33]. The visual representation of estrogen receptor signalling pathway is given below in Fig. 1.

3.2 Progesterone signaling in breast cancer

Progesterone signaling in BC is mediated primarily through the PR which exists in two isoforms, PR-A and PR-B and operates via both genomic and non-genomic pathways. In the classical genomic pathway, progesterone binds to nuclear PRs with PR-B playing a dominant role in promoting proliferation by regulating genes such as cyclin D1 which drives cell cycle progression [34, 35]. PR-A, while less transcriptionally active, modulates

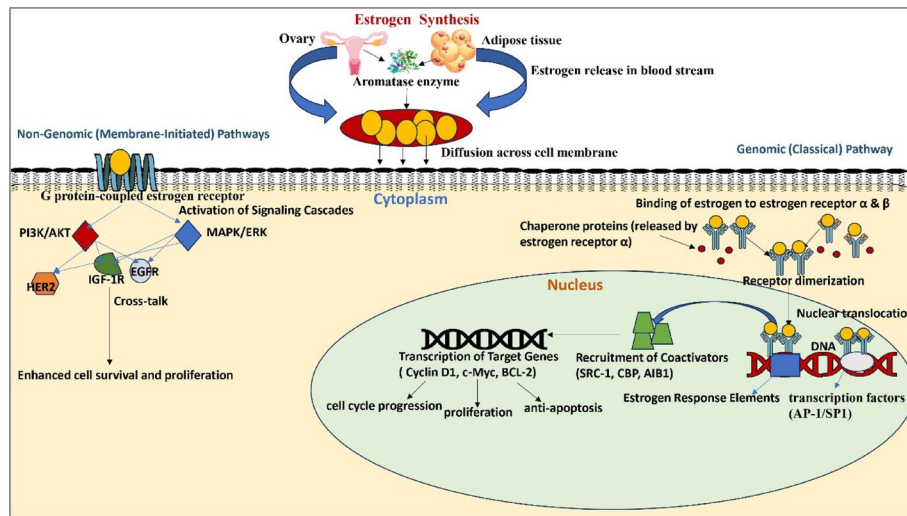


Fig. 1 Mechanism of estrogen receptor signalling for breast cancer progression

specific gene sets and an imbalance favoring PR-A over PR-B is associated with altered mammary gland architecture, ductal hyperplasia and increased cancer risk [36]. Non-genomic signaling involves rapid activation of kinase pathways (e.g., PI3K/AKT, ERK) by PR-B, enhancing cell survival and proliferation [37]. Progesterone also exerts paracrine effects in normal breast tissue where PR-positive cells secrete factors like RANKL and WNT4, stimulating neighboring PR-negative cells to proliferate via the RANKL/RANK/NF- κ B and WNT/ β -catenin pathways, respectively [38, 39]. In BC, this can shift to auto-crine signaling, enabling self-sustained tumor growth, particularly in BRCA1-mutant tumors where RANKL/RANK signaling promotes stemness and tumorigenesis [35, 37, 40]. Additionally, progesterone upregulates serum/glucocorticoid-regulated kinase 1 by downregulating microRNAs (e.g., miR-29a, miR-101-1), activating the serum/glucocorticoid-regulated kinase 1/NDRG1 axis, which inhibits kinases like AKT1 and ERK1/2, reducing cell invasion and migration in both PR-positive and PR-negative BC via glucocorticoid receptor mediation [41]. Progesterone also modulates ER activity by redirecting ER α chromatin binding, altering gene expression to suppress estrogen-driven proliferation in ER+/PR+ tumors, an effect more pronounced in high-PR luminal A subtypes. The interplay with pathways like Notch further supports CSC maintenance, contributing to tumor aggressiveness [42].

3.3 HER2 signalling pathway

The HER2 signalling pathway drives BC progression through a cascade initiated by ERBB2 gene amplification on chromosome 17q12, leading to 10-100-fold overexpression of the 185-kDa HER2 receptor tyrosine kinase which lacks a known ligand and adopts a constitutively active conformation for dimerization, unlike other EGFR family members (EGFR, HER3, HER4). This overexpression promotes spontaneous homodimerization (HER2:HER2) or heterodimerization, particularly with HER3 whose inactive kinase domain enhances oncogenic signaling via extracellular domain interactions stabilized by disulfide bonds and hydrophobic interactions [43]. Dimerization triggers trans-autophosphorylation of tyrosine residues (e.g., Y1139, Y1196, Y1248) on HER2 or HER3's intracellular tail, creating docking sites for adaptor proteins with SH2 or PTB domains

[44]. Grb2 and Shc bind Y1139 to activate the RAS/MAPK pathway, while p85 (PI3K's regulatory subunit) binds phosphorylated HER3, amplifying the PI3K/AKT/mTOR pathway [45]. In the PI3K/AKT/mTOR pathway, p85 activates p110, converting phosphatidylinositol 3,4,5-diphosphate to phosphatidylinositol 3,4,5-trisphosphate which recruits AKT for phosphorylation at Thr308 (by phosphoinositide-dependent kinase 1) and Ser473 (by mTOR complex 2), fully activating AKT to inhibit pro-apoptotic proteins (BAD, caspase-9), upregulate anti-apoptotic Bcl-2 and activate mTOR complex 1 which phosphorylates S6K1 and 4EBP1 to enhance protein translation, ribosome biogenesis and glycolysis (Warburg effect), with *PIK3CA* mutations or PTEN loss (30% of cases) amplifying this pathway for aggressive growth and therapy resistance [46]. Concurrently, Grb2 recruits SOS to activate RAS (e.g., KRAS, NRAS) by exchanging GDP for GTP, leading to RAF/MEK/ERK activation, where ERK phosphorylates transcription factors (ELK1, c-Jun, c-Fos) to upregulate CCND1, c-Myc (for proliferation) and VEGF (for angiogenesis) while promoting invasion via cytoskeletal remodeling and matrix metalloproteinase expression [47]. In HR+/HER2 + tumors (50%), HER2-driven PI3K/AKT signaling phosphorylates ER at Ser167/118 for ligand-independent activation of CCND1 and MYC while ER upregulates HER2 via AP-2 and forkhead box O3a, forming a feedback loop that enhances growth and endocrine therapy resistance [48]. HER2 also activates phospholipase C gamma (for motility via calcium/PKC), STAT5 (for survival and cytokine production) and NF- κ B (for inflammation and apoptosis resistance) [46]. These pathways drive proliferation (CCND1, c-Myc), survival (Bcl-2, Bcl-XL), invasion/metastasis (EMT genes SNAIL, SLUG, VEGF), cancer stem cell maintenance (Notch, Wnt/ β -catenin) and immune evasion (PD-L1, HLA-G, IL-6-driven Treg/M2 macrophage recruitment) [48]. Resistance emerges from p95HER2 (in 30% of cases), a truncated HER2 evading trastuzumab but sensitive to TKIs like lapatinib, HER2 mutations (L755S, T798M in 2–5%), compensatory pathways (EGFR, HER3, IGF-1R, MET), *PIK3CA* mutations/PTEN loss (20–30%), TME-driven immune evasion (PD-L1/HLA-G in 20%), and hypoxia/TGF- β , all sustaining PI3K/AKT or MAPK signalling and reducing trastuzumab/endocrine therapy efficacy [49]. The visual representation of the HER2 signalling pathway is given in Fig. 2.

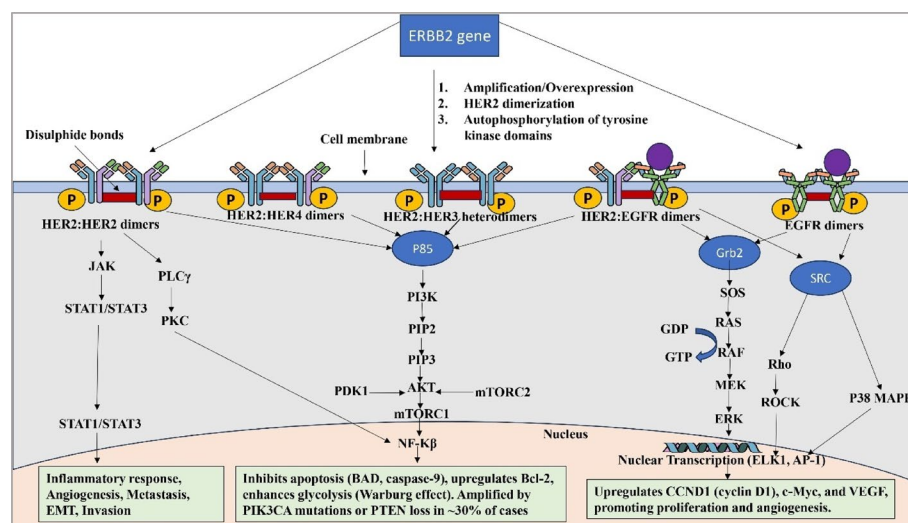


Fig. 2 HER2 signalling pathway for breast cancer progression

3.4 PI3K/AKT/mTOR pathway

The PI3K/AKT/mTOR pathway is a central regulator of cell growth, proliferation, survival and metabolism, frequently dysregulated in BC particularly in HR + and HER2 + subtypes [50]. Activation begins when growth factor receptors (e.g., HER2, EGFR) or other stimuli activate PI3K which phosphorylates phosphatidylinositol 4,5-bisphosphate to phosphatidylinositol 3,4,5-trisphosphate. Phosphatidylinositol 3,4,5-trisphosphate recruits AKT to the cell membrane where it is phosphorylated and activated by phosphoinositide-dependent kinase 1 and mTOR complex 2. Activated AKT stimulates mTOR complex 1 which regulates protein synthesis via S6K1 and 4EBP1, promotes cell cycle progression and inhibits apoptosis by inactivating pro-apoptotic proteins like BAD and activating anti-apoptotic signals such as Bcl-2 [51, 52]. In BC, PIK3CA mutations encoding the catalytic subunit of PI3K occur in ~ 40% of HR + tumors, leading to constitutive pathway activation, enhanced glycolysis (Warburg effect) and resistance to therapies [53]. This pathway also supports tumor metabolism by increasing glucose uptake and lipid synthesis, critical for rapidly proliferating cancer cells [54].

3.5 RAS/MAPK pathway

3.5.1 Mechanism

The RAS/MAPK pathway regulates cell proliferation, differentiation, and survival, and is frequently dysregulated in BC, particularly in HER2 + and TNBC subtypes [54, 55]. The cascade is initiated when growth factors (e.g., EGF, TGF- α) bind receptor tyrosine kinases like HER2 or EGFR, leading to receptor dimerization and autophosphorylation. Adaptor proteins (e.g., Grb2) recruit SOS, which activates RAS by exchanging GDP for GTP. Activated RAS recruits RAF (e.g., BRAF, CRAF), which phosphorylates MEK, subsequently activating ERK. ERK translocates to the nucleus, regulating transcription factors (e.g., ELK1, AP-1) to promote cyclin expression and inhibit CDK inhibitors, driving cell cycle progression [56]. In HER2 + BC, MAPK signaling enhances tumor growth by upregulating anti-apoptotic proteins (e.g., BCL-2) and promoting angiogenesis and invasion via ERK-mediated motility [57].

3.6 Wnt/ β -catenin signaling pathway

The Wnt/ β -catenin pathway is critical for cell fate determination, EMT, and CSC maintenance in BC, particularly in TNBC [58]. In the absence of Wnt ligands, β -catenin is phosphorylated by a destruction complex (APC, GSK-3 β , CK1 α), targeting it for degradation. When Wnt ligands bind Frizzled receptors and LRP co-receptors, GSK-3 β is inhibited, stabilizing β -catenin, which translocates to the nucleus and interacts with T-Cell Factor/Lymphoid Enhancer Factor transcription factors to activate oncogenic genes (e.g., c-Myc, cyclin D1, survivin) [59]. In BC, pathway activation occurs via mutations in APC or CTNNB1 (β -catenin), epigenetic silencing of inhibitors (e.g., SFRP1, DKK1), or stromal cell-derived Wnt ligands, promoting autocrine and paracrine signaling [60]. This drives tumor growth, metastasis, and therapy resistance, particularly in aggressive subtypes.

3.7 JAK/STAT signaling pathway

The JAK/STAT pathway mediates cytokine signaling, immune response and tumor immune evasion, particularly in TNBC and inflammatory BC [61]. Cytokines (e.g.,

IL-6) bind receptors, activating JAK kinases (JAK1, JAK2, TYK2) which phosphorylate STAT proteins (e.g., STAT3, STAT5). Phosphorylated STATs dimerize, translocate to the nucleus and regulate genes promoting proliferation, survival and EMT (e.g., Bcl-XL) [62]. In BC, STAT3/5 hyperactivation, driven by cytokine overproduction or epigenetic suppression of negative regulators (e.g., SOCS3), enhances angiogenesis, CSC maintenance and immune suppression [63]. Crosstalk with HER2 and ER pathways exacerbates resistance in HR + and HER2 + tumors [64].

3.8 Notch signaling pathway

The Notch pathway regulates cell differentiation, stemness and EMT, with significant implications in BC progression and resistance [65]. Notch signaling involves four receptors (Notch1-4) and five ligands (Jagged1/2, Delta-Like Ligand 1/3/4). Ligand-receptor binding triggers proteolytic cleavage, releasing the Notch intracellular domain which translocates to the nucleus and interacts with CSL/RBP-J κ to activate genes (e.g., Hes, Hey) promoting EMT, metastasis and CSC maintenance [66]. In BC, Notch1/4 and Jagged1 overexpression correlate with poor prognosis, particularly in TNBC and HER2 + subtypes where Notch signaling enhances angiogenesis and drug resistance via interactions with PI3K/AKT and NF- κ B [67].

4 Therapeutic implications of breast cancer subtypes

Breast cancer's molecular subtypes drive distinct therapeutic strategies due to their varied treatment responses. Luminal subtypes, often HR + respond well to endocrine therapies like tamoxifen, HER2-enriched tumors benefit from targeted therapies such as trastuzumab, while basal-like cancers, often triple-negative pose challenges due to limited targeted options, relying on chemotherapy [24, 25]. Research findings underscore the need for precision medicine, tailoring treatments to subtype-specific molecular profiles to optimize outcomes and address resistance, paving the way for novel therapies in personalized breast cancer management. Table 1 summarizes the therapeutic implications, key therapies, clinical evidence and targeted nano-formulations for the Luminal A, Luminal B, HER2-Enriched, and TNBC subtypes.

5 Therapeutic resistance to signalling pathways

5.1 Estrogen receptor

Endocrine therapies (SERMs, SERDs, AIs) targeting ER signalling in ER + BC often face intrinsic or acquired resistance through dynamic interactions with multiple mechanisms: ESR1 mutations (Y537S, D538G, 20–40% of metastatic cases) cause ligand-independent ER activation, sustaining CCND1 and MYC transcription and reducing SERM/AI efficacy [105]; alternative pathways like PI3K/AKT/mTOR (altered in 30–40% via PIK3CA mutations/PTEN loss) or amplified HER2/FGFR1-MAPK/ERK signalling bypass ER dependency, with bidirectional crosstalk phosphorylating ER or upregulating growth factor receptors to maintain tumor growth [106]; epigenetic changes including ER target gene promoter hypermethylation or NCOR1 upregulation silence ER-dependent genes, shifting reliance to alternative pathways and reinforcing resistance via histone deacetylase feedback loops [6]; the TME via stromal/immune interactions and cytokine signalling activates integrin/PI3K/AKT/YAP/TAZ pathways, supported by non-genomic ER

signaling through G protein-coupled ER/membrane-bound ERs, providing proliferative signals despite genomic ER inhibition [6]; CDK4/6 inhibitor resistance arises from retinoblastoma 1 loss, CCNE1 amplification or S6K1-driven c-Myc activation, bypassing the G1/S checkpoint and reducing ER dependency [107, 108]; upregulated BCL-2 enhances survival and CSC enrichment promotes recurrence, both linked to ER signaling; therapeutic strategies include dual ER/PI3K/AKT/mTOR targeting with alpelisib/everolimus, combining CDK4/6 inhibitors (palbociclib) with SERDs/AIs, next-generation SERDs/PROTACs for ESR1 mutants, HDAC inhibitors to restore ER sensitivity and CSC/micro-environment-targeted approaches to disrupt resistance networks and improve outcomes [106].

5.2 Progesterone receptor

Resistance to therapies targeting progesterone signaling is a significant challenge driven by PR isoform imbalance, pathway crosstalk and stem cell expansion. A high PR-A/PR-B ratio, common in ductal carcinoma in situ and invasive BC promotes endocrine resistance, particularly to tamoxifen and predicts poorer prognosis due to increased expression of growth factors like amphiregulin which fuel paracrine-driven tumor growth [109]. RANKL/RANK signalling contributes to therapy resistance by enhancing CSC populations and mesenchymal phenotypes particularly in ER+/HER2 – and BRCA1-mutant BC, leading to increased invasiveness and metastasis [110, 111]. Compensatory activation of pathways like PI3K/AKT or MAPK often triggered by progesterone-induced NF- κ B signaling further drives resistance to endocrine and targeted therapies. To overcome these challenges, dual inhibition of NF- κ B and USF-1 could disrupt progesterone-driven tumorigenesis, reducing stemness and restoring therapy sensitivity [112–115]. Combining RANKL inhibitors (e.g., denosumab) with endocrine therapy or chemotherapy enhances tumor cell differentiation and immune response, countering resistance [116, 117]. Targeting the serum/glucocorticoid-regulated kinase 1/NDRG1 axis with Serum/Glucocorticoid-Regulated Kinase 1 activators or inhibitors of downstream kinases (e.g., AKT1, ERK1/2) may reduce metastatic potential, even in PR-negative BC [118]. Additionally, integrating progesterone-based therapies with PI3K/AKT or MEK inhibitors addresses compensatory pathway activation, while epigenetic strategies to restore PR expression in TNBC could sensitize tumors to progesterone's anti-invasive effects [113, 114]. Molecular profiling to assess PR isoform ratios, RANK expression, and Serum/Glucocorticoid-Regulated Kinase 1 levels is essential for designing personalized treatments to overcome resistance and improve outcomes [119, 120].

5.3 HER2

Resistance to HER2-targeted therapies in HER2-positive BC arises from multiple mechanisms: truncated p95HER2 (up to 30% of cases) lacks the extracellular domain, evading trastuzumab/pertuzumab but retaining kinase activity and is sensitive to TKIs like lapatinib/neratinib, sustaining PI3K/AKT/MAPK signaling [121, 122]; HER2 mutations (e.g., L755S, T798M, 2–5% prevalence) reduce antibody/TKI binding, stabilizing active kinase conformations and requiring agents like T-DXd [123, 124]; tumor heterogeneity and dynamic HER2 downregulation reduce antibody/ADC efficacy, necessitating TKIs or ADCs with bystander effects (e.g., T-DXd), with liquid biopsies monitoring resistance [125–127]; compensatory pathways like EGFR/HER3 (HER3 overexpressed in 50% of

Table 1 Therapeutic implications of breast cancer subtypes, key therapies, clinical evidence and targeted nano-formulations

Subtype	Therapeutic Implications	Key Therapies	Clinical Evidence	Targeted Nanoformulations	References
Luminal A	(i) Endocrine therapies target high ER/PR expression, effective for hormone-dependent tumors with low proliferation rates; limited in ESR1-mutated tumors. (ii) Progesterone-based therapies combine with ER antagonists to inhibit estrogen-driven growth via PR modulation of ERα chromatin binding. (iii) RANKL inhibitors target RANKL/RANK/NF-κB axis to enhance immune responses and sensitivity to endocrine therapies. (iv) Resistance mechanisms: ESR1 mutations (Y537S, D538G) cause ligand-independent ER activation; epigenetic silencing of ER target genes; high PR-A/PR-B ratio promotes resistance to tamoxifen. (v) Solutions: Next-generation SERDs (e.g., camizestrant) for ESR1 mutants; HDAC inhibitors to restore ER sensitivity; dual ER/PI3K targeting for compensatory pathways.	(i) SERMs: Tamoxifen, Toremifene, Raloxifene (competitive ER binding, block proliferative gene transcription, FDA-approved for early/advanced ER + BC, limited in ESR1 mutants). (ii) AIs: Anastrozole, Letrozole, Exemestane (inhibit estrogen synthesis, effective in postmenopausal women, less effective in premenopausal due to ovarian estrogen). (iii) SERDs: Fulvestrant (induces ER degradation, standard for advanced ER + BC). (iv) Progesterone/Progestins: Modulate ER activity, reprogram ERα binding for better prognosis. (v) RANKL inhibitors: Denosumab (enhances immune response, sensitizes to endocrine therapies).	(i) 5-year survival rates >90% due to high response to endocrine therapy. (ii) Effective in early and advanced ER + BC, though limited in ESR1-mutated tumors.	(i) Polyethylene glycolylated chitosan nanoparticles with monoclonal antibodies deliver doxorubicin for targeted ER inhibition. (ii) Folate-targeted curcumin-loaded niosomes modulate NF-κB. (iii) Injectable albumin nanoparticles for sustained chemotherapy release. (iv) PLGA-based nanostructures for raloxifene delivery to enhance bioavailability and target ER + tumors.	[1, 9, 12, 68–74]

Table 1 (continued)

Subtype	Therapeutic Implications	Key Therapies	Clinical Evidence	Targeted Nanoformulations	References
Luminal B	(i) Combination therapies address higher proliferation (elevated Ki-67) and HER2/EGFR co-expression; endocrine therapies with CDK4/6 inhibitors; PI3K/AKT/mTOR inhibitors for PIK3CA-mutant or AI-resistant tumors; dual ER/HER2 targeting for triple-positive BC (TPBC). (ii) Next-generation therapies: SERDs/PROTACs for ESR1 mutants; HDAC/BCL-2 inhibitors to restore sensitivity; RANKL/NF-κB/USF-1 inhibitors for resistance; ADCs for low HER2 expression. (iii) Personalized approaches: Multi-omics and AI-driven biomarker discovery; target CSCs and TME. (iv) Resistance mechanisms: ESR1 mutations (20–40% in metastatic cases); PI3K/AKT/mTOR activation (PIK3CA mutations in ~40%); crosstalk with HER2/FGFR1-MAPK/ERK; Retinoblastoma 1 loss or CCNE1 amplification for CDK4/6 resistance; Treg/CAF interactions in TME. (v) Solutions: Dual ER/PI3K targeting (alpelisib + fulvestrant); CDK4/6 inhibitors with SERDs; Notch/Wnt inhibitors for CSCs; anti- Cytotoxic T-Lymphocyte-Associated Protein 4for Treg depletion.	(i) SERMs: Tamoxifen, Toremifene, Raloxifene. (ii) AIs: Anastrozole, Letrozole, Exemestane. (iii) SERDs: Fulvestrant, Elacestrant (FDA-approved for ESR1-mutated ER+/HER2-). (iv) PROTACs: Vepdegestrant/ARV-471 (degrades ER mutants). (v) Selective Estrogen Receptor Covalent Antagonists: Block/inhibit ER Irreversibly. (vi) CDK4/6 inhibitors: Palbociclib, Ribociclib, Abemaciclib (block G1-S transition). (vii) PI3K inhibitors: Alpelisib (PIK3CA mutants). (viii) mTOR inhibitors: Everolimus. (ix) EGFR/HER2 inhibitors: Lapatinib. (x) HDAC inhibitors: Entinostat. (xi) BCL-2 inhibitors: Venetoclax. (xii) HER2 therapies: Trastuzumab, Pertuzumab. (xiii) ADCs: Sacituzumab Govitecan, T-DXd. (xiv) RANKL inhibitors: Denosumab. (xv) NF-κB/USF-1 inhibitors.	(i) monarchE (NCT03155997): Abemaciclib + endocrine therapy reduced recurrence risk by 30% in high-risk, node-positive early-stage disease. (ii) PALLAS (NCT02513394): Palbociclib + endocrine therapy showed no significant invasive disease-free survival benefit. (i) MONALEESA-2 (NCT01958021): Ribociclib + Letrozole extended PFS to 25.3 months vs. 16.0 months. (ii) PERTAIN (NCT01491737): Pertuzumab + Trastuzumab + AI improved PFS to 18.9 months vs. 15.8 months in HR+/HER2 + metastatic disease. (iii) SOLAR-1 (NCT02437318): Alpelisib extended PFS to 11.0 months vs. 5.7 months in PIK3CA-mutated HR+/HER2- advanced BC. (iv) EMERALD: Elacestrant reduced progression/death risk by 45% in ESR1-mutated ER+/HER2- advanced cases. (v) VERITAC-2: Vepdegestrant improved PFS to 5 months vs. 2.1 months for Fulvestrant in ESR1-mutated ER+/HER2- BC. (vi) DESTINY-Breast06 (NCT04494425): Ongoing trial evaluating T-DXd in HR+/HER2-low disease. (vii) B-PRECISE-01 (NCT03767335): Evaluates PI3K inhibitors in TPBC. (viii) Phase Ib trial: Entinostat + Nivolumab + Ipilimumab improved response rates in advanced HER2-negative BC.	(i) Polyethylene glycolylated chitosan nanoparticles for doxorubicin. (ii) Folate-targeted nanoparticles for curcumin. (iii) PLGA-based nanostructures for raloxifene/doxorubicin. (iv) siRNA-chitosan nanoformulations targeting MicroRNA – 21/AKT/ERK. (v) Cisplatin-loaded chitosan nanoparticles with rituximab for targeted cytotoxicity.	[1, 2, 7, 8, 9, 11, 12, 15, 68–84]

Table 1 (continued)

Subtype	Therapeutic Implications	Key Therapies	Clinical Evidence	Targeted Nanoformulations	References
HER2-Enriched	(i) Target ERBB2 amplification with monoclonal antibodies, ADCs, and TKIs, effective in HER2-low tumors. (ii) Combination therapies with PI3K/AKT inhibitors, ICIs, or bispecific antibodies to overcome resistance; Notch inhibitors for CSC-driven resistance; AXL/MET inhibitors for pathway crosstalk; exosome inhibitors for resistance signaling. (iii) Personalized approaches: Multi-omics for biomarker-driven therapy; oncolytic viruses and Major Histocompatibility Complex class I agonists to enhance immune recognition. (iv) Resistance mechanisms: Truncated p95HER2 (30%) evades antibodies; HER2 mutations (L755S, T798M, 2–5%); tumor heterogeneity; EGFR/HER3/MET/IGF-1R activation; PIK3CA mutations/PTEN loss (20–30%); TME factors (HLA-G, PD-L1, TGF-β); ER cross-talk in HR+/HER2+ tumors. (v) Solutions: ADCs (T-DXd) for HER2-low/mutant tumors; dual HER2 blockade (trastuzumab + pertuzumab/TKIs); PI3K inhibitors (alpelisib); ICIs (pembrolizumab); Notch/AXL/MET inhibitors; PROTACs; liquid biopsies for resistance monitoring.	(i) Monoclonal antibodies: Trastuzumab (blocks homodimerization, induces ADCC), Pertuzumab (blocks HER2/HER3), Margetuximab (enhanced ADCC). (ii) ADCs: T-DM1 (trastuzumab + DM1), T-DXd (topoisomerase I inhibitor, bystander killing). (iii) TKIs: Lapatinib, Neratinib, Tucatinib (targets brain metastases). (iv) PI3K inhibitors: Alpelisib. (v) ICIs: Pembrolizumab, Durvalumab. (vi) Bispecific antibodies: KN026, Zanidatamab. (vii) Notch inhibitors: Tarextumab, anti-Delta-Like Ligand. (viii) AXL/MET inhibitors. (ix) Exosome inhibitors: Annexin V. (x) PROTACs. (xi) Oncolytic viruses. (xii) Major Histocompatibility Complex class I agonists.	(i)CLEOPATRA (NCT00567190): Trastuzumab + Pertuzumab + Docetaxel extended PFS to 18.5 months vs. 12.4 months. (ii)DESTINY-Breast03 (NCT03529110): T-DXd achieved 75.8% 12-month PFS vs. 34.1% with T-DM1. (iii)APHINITY (NCT01358877): Adjuvant Pertuzumab + Trastuzumab reduced recurrence risk by 19% in node-positive disease. (iv)DESTINY-Breast04 (NCT03734029): T-DXd improved PFS to 10.1 months vs. 5.4 months in HR+/HER2-low metastatic disease. (v)HER2CLIMB (NCT02614794): Tucatinib + Trastuzumab + Capecitabine extended PFS to 7.8 months vs. 5.6 months. (vi)ExteNET: Neratinib reduced recurrence in early-stage, node-positive disease. (vii)EMILIA: T-DM1 effective in residual/metastatic disease post-trastuzumab. (viii)SOPHIA: Margetuximab improved PFS in low-affinity Fcγ receptor genotypes post-trastuzumab. (ix)NCT04784715: Ongoing trial exploring T-DXd with Pertuzumab or Durvalumab. (x)DESTINY-Breast06 (NCT0494425): Ongoing trial evaluating T-DXd in HR+/HER2-low disease. (xi)B-PRECISE-01 (NCT03767335): Evaluates PI3K inhibitors in TPBC.	(i) Liposomes/PLGA nanoparticles for T-DM1/T-DXd delivery. (ii) Chitosan nanoparticles for cisplatin + rituximab. (iii) Polyphenol-loaded nanocarriers (curcumin, quercetin) targeting NF-κB. (iv) Nanostructured lipid carriers for ellagic acid. (v) Nanoparticle-based systems for co-delivery of HER2 inhibitors and immunotherapeutics.	[8, 9, 11, 68, 75, 84–93]

Table 1 (continued)

Subtype	Therapeutic Implications	Key Therapies	Clinical Evidence	Targeted Nanoformulations	References
TNBC	(i) PARP inhibitors for BRCA1/2 mutants exploit Homologous Recombination Repair defects. (ii) ICIs for PD-L1 + tumors to enhance anti-tumor immunity. (iii) ADCs targeting TROP2; AR inhibitors for LAR subtype. (iii) Additional therapies: Glutaminase inhibitors, Notch/JAK/STAT inhibitors for resistance; chemotherapy combinations; epigenetic therapies (HDAC inhibitors) to sensitize tumors; TGF-β inhibitors for immune modulation; exosome/VEGF inhibitors for metastasis; fasting-mimicking diets to reduce CSCs; nanoparticle-based delivery; cancer vaccines; Chk1/MEK/PI3K inhibitors; RNA-based therapies; ATR inhibitors for ATM-deficient tumors. (iv) Personalized approaches: Multi-omics and AI-driven therapy; combination with radiotherapy or oncolytic viruses. (v) Resistance mechanisms: Immune evasion (Tregs, Myeloid-Derived Suppressor Cells, PD-L1); DNA repair restoration in BRCA1/2 mutants; ABC transporter overexpression; PI3K/AKT, JAK/STAT, Notch activation; EMT; CSC enrichment. (vi) Solutions: Combine PARP inhibitors with ICIs; Notch inhibitors (anti-Notch1 / Delta-Like Ligand); JAK/STAT inhibitors; anti-VEGF antibodies; nanoparticle delivery to bypass efflux; epigenetic therapies to restore sensitivity; TGF-β inhibitors to reduce Tregs.	(i) PARP inhibitors: Olaparib, Talazoparib (synthetic lethality for BRCA1/2 mutants). (ii) ICIs: Pembrolizumab, Atezolizumab, Nivolumab, Ipilimumab. (iv) ADCs: Sacituzumab Govitecan (TROP2-targeted). (iv) AR inhibitors: Enzalutamide (LAR subtype). (v) Glutaminase inhibitors: CB-839. (vi) Notch inhibitors: Tarextumab, anti-Delta-Like Ligand, Hesi1/Hey1-targeting. (vi) JAK/STAT inhibitors: Ruxolitinib, Napabucasin. (vii) Chemotherapy: Anthracyclines, Taxanes, Carboplatin. (ix) HDAC inhibitors: Entinostat, Vorinostat. (x) TGF-β inhibitors: Galunisertib. (xi) Exosome inhibitors: GW4869. (xii) VEGF inhibitors: Bevacizumab. (xiii) Chk1/MEK/PI3K inhibitors: Ipatasertib. (xiv) RNA-based therapies: siRNAs (MYC, SNAIL). (xv) PROTACs. (xvi) ATR inhibitors: Ceralasertib. (xvii) Nanoparticle-based systems: Chitosan, PLGA-based. (xviii) Cancer vaccines: MUC-1, NY-ESO-1.	(i) IMpassion130 (NCT02425891): Atezolizumab + nab-Paclitaxel improved PFS to 7.2 months vs. 5.5 months in PD-L1 + metastatic TNBC. (ii) OlympiAD (NCT02000622): Olaparib extended PFS to 7.0 months vs. 4.2 months in BRCA-mutated disease. (iii) EMBRACA (NCT01945775): Talazoparib extended PFS to 8.6 months vs. 5.6 months. (iv) FUTURE (NCT03805399): Genomic profiling-guided therapies (e.g., Everolimus, Camrelizumab) achieved 28.1% ORR. (v) ASCENT (NCT02574455): Sacituzumab Govitecan extended PFS to 5.6 months vs. 1.7 months. (vi) NCT01889238: Enzalutamide showed 29% clinical benefit rate at 24 weeks in LAR subtype. (vii) GeparSixto (NCT01426880): Carboplatin + neoadjuvant chemotherapy increased Pathological Complete Response to 53.2%. (viii) NCT02299999: Ongoing trial evaluating Ipatasertib. (ix) Phase Ib trial: Entinostat + Nivolumab + Ipilimumab improved response rates in advanced HER2-negative BC.	(i) siRNA-chitosan nanoformulations targeting MicroRNA – 21/AKT/ERK. (ii) Cisplatin-loaded chitosan nanoparticles with rituximab. (iii) PLGA-based nanostructures for doxorubicin. (iv) Polyphenol-loaded nanocarriers (curcumin, quercetin) targeting NF-κB. (v) Nanostructured lipid carriers for ellagic acid. (vi) Nanoparticle-based systems for co-delivery of chemotherapy/immunotherapeutics. (vii) BC vaccines with nanostructured lipid carriers.	[1, 5, 8, 10, 11, 15, 68, 69, 75, 82–86, 88, 90–104]

resistant cases), IGF-1R or MET (amplified in 5–10%) sustain PI3K/AKT/MAPK signaling, while *PIK3CA* mutations/PTEN loss (20–30%) maintain mTOR/AKT activity, reducing trastuzumab efficacy and requiring PI3K inhibitors (e.g., alpelisib) [128–131]; Src/integrin signalling and genomic instability further amplify bypass mechanisms [129, 132]; the TME contributes via HLA-G overexpression (20% of cases), inhibiting NK cell-mediated ADCC, PD-L1 upregulation (10–20%), suppressing T-cells and macrophage/TGF- β -driven immune evasion, with hypoxia/hypoxia-inducible factor 1 α enhancing survival pathways, addressed by T-DXd and PD-1/PD-L1 inhibitors (e.g., pembrolizumab) [121, 132–135]; in HR+/HER2 + tumors (50% of cases), HER2-driven PI3K/AKT/mTOR phosphorylates ER (Ser167/118), enabling ligand-independent transcription while ER upregulates HER2 via AP-2/ Forkhead Box O3a, forming a feedback loop that sustains resistance, with shared PI3K/AKT/mTOR activation exacerbating it, as seen in the TAnDEM trial where trastuzumab-anastrozole improved outcomes, necessitating dual blockade with HER2 agents, CDK4/6 inhibitors (e.g., palbociclib) or PI3K inhibitors guided by molecular profiling [128, 136–140].

5.4 PI3K/AKT/mTOR

Resistance to PI3K/AKT/mTOR inhibitors arises through multiple mechanisms. Compensatory activation of upstream receptors (e.g., HER2, EGFR) or alternative pathways (e.g., RAS/MAPK, JAK/STAT) restores oncogenic signalling [50, 141]. Feedback loops such as mTOR complex 1 inhibition leading to AKT reactivation via mTOR complex 2 reduce inhibitor efficacy [53]. PTEN loss or mutations in *PIK3CA* can sustain pathway activation, while upregulation of drug efflux pumps (e.g., ABC transporters) reduces drug intracellular concentrations [52]. In TNBC, AKT activation or PTEN deletion further complicates therapy response [142]. Strategies to overcome resistance include combining PI3K inhibitors with MEK inhibitors (e.g., trametinib) to block MAPK cross-talk or using CDK4/6 inhibitors to target cell cycle progression [7]. Biomarker-driven approaches such as *PIK3CA* mutation status guide patient selection to improve outcomes [143].

5.5 RAS/MAPK

Resistance to RAS/MAPK-targeted therapies often results from secondary mutations in KRAS, NRAS or BRAF which restore pathway activation. Crosstalk with PI3K/AKT/mTOR or JAK/STAT pathways provides alternative survival signals, reducing inhibitor efficacy [13]. EMT, driven by ERK activation, promotes a drug-resistant, invasive phenotype particularly in HER2 + BC. Upregulation of receptor tyrosine kinases (e.g., MET, FGFR) or feedback activation of upstream receptors also contributes to resistance [144–146]. Combining MEK inhibitors with HER2-targeted therapies or PI3K inhibitors shows promise in preclinical studies by targeting convergent downstream effectors [70]. Molecular profiling of RAS/MAPK alterations is critical to tailor therapies and mitigate resistance [55].

5.6 Wnt/ β -catenin

Resistance to Wnt inhibitors arises from compensatory pathway activation (e.g., PI3K/AKT, Notch) or upregulation of alternative Wnt ligands [58]. Epigenetic modifications such as promoter hypermethylation of Wnt antagonists sustain pathway activity [59].

CSC enrichment, driven by β -catenin-mediated stemness further promotes resistance to chemotherapy and targeted therapies [14]. Combining Wnt inhibitors with ICIs or MEK inhibitors may overcome resistance by targeting TME interactions and parallel pathways [58]. Ongoing research focuses on identifying biomarkers to predict response and optimize combination strategies [14].

5.7 Notch signaling

Resistance to Notch inhibitors arises from compensatory activation of PI3K/AKT, MAPK or Wnt pathways which sustain tumor survival [82]. Upregulation of drug efflux pumps and CSC enrichment via Notch-mediated stemness further reduce efficacy [65]. Toxicity from non-specific Notch inhibition due to its role in normal tissue homeostasis, limits clinical use [82]. Combining Notch inhibitors with ICIs or PI3K inhibitors may overcome resistance by targeting TME and parallel pathways [82]. Molecular profiling of Notch expression is critical for personalized therapy [67].

6 Pathway interactions and strategies to overcome crosstalk-driven resistance

The PI3K/AKT/mTOR and RAS/MAPK pathways exhibit significant crosstalk, driving resistance in BC through shared downstream effectors and feedback loops [13, 25]. In HR + BC, PIK3CA mutations activate AK which upregulates MAPK signaling via receptor tyrosine kinase crosstalk, bypassing endocrine therapy [7, 139]. Similarly, in HER2 + BC, RAS/MAPK activation via HER2 or EGFR sustains tumor growth despite PI3K inhibition [145]. Recent studies show that PI3K inhibition can trigger MAPK reactivation through mTOR complex 2-mediated feedback, reducing Alpelisib efficacy [140, 141]. Saini et al. (2013) demonstrated that dual inhibition of PI3K and MAPK pathways restores sensitivity in PIK3CA-mutant tumors by targeting convergent effectors like cyclin D1 [13]. In TNBC, MAPK hyperactivation compensates for PI3K inhibition, promoting EMT and metastasis [143]. Tiu-Lim et al. (2021) identified RAS/MAPK alterations in metastatic BC, correlating with resistance to HER2-targeted therapies, suggesting dual PI3K/MEK inhibition as a strategy [145]. The Wnt/ β -catenin pathway interacts with PI3K/AKT to enhance CSC maintenance and resistance. In TNBC, Wnt signaling upregulates PI3K/AKT via GSK-3 β inhibition, sustaining β -catenin activity and stemness. Zhang et al. (2023) showed that Wnt inhibitors combined with PI3K inhibitors reduce CSC populations, overcoming resistance [14]. JAK/STAT signaling also crosstalks with PI3K/AKT, particularly in TNBC where IL-6-driven STAT3 activation enhances AKT signaling, promoting immune evasion [62, 140]. Manore et al. (2022) highlighted that JAK/STAT inhibitors combined with PI3K inhibitors improve outcomes in TNBC by blocking cytokine-mediated resistance [140]. Notch signaling interacts with PI3K/AKT and NF- κ B to drive resistance in TNBC and HER2 + BC. Krishna et al. (2019) reported that notch1 activation upregulates PI3K/AKT, sustaining CSC populations and resistance to trastuzumab [82]. Combining Notch inhibitors with PI3K or HER2 inhibitors enhances sensitivity by targeting these interactions. Edwards and Brennan (2021) noted that Notch-driven EMT promotes a mesenchymal phenotype, reducing chemotherapy efficacy which can be countered by dual Notch/PI3K inhibition [65].

Combination therapies targeting pathway crosstalk are critical for overcoming resistance in BC. Dual PI3K/MEK inhibition such as Alpelisib combined with trametinib targets shared effectors in the PI3K/AKT/mTOR and MAPK pathways, reducing resistance

in PIK3CA-mutant HR + BC [13, 143]. Similarly, combining CDK4/6 inhibitors (e.g., palbociclib) with PI3K inhibitors addresses cell cycle-driven resistance in preclinical models, enhancing antiproliferative effects [140]. In HER2-positive BC, integrating MEK inhibitors with trastuzumab overcomes MAPK-mediated resistance, improving therapeutic outcomes [70]. For TNBC, combining JAK/STAT inhibitors (e.g., ruxolitinib) with ICIs enhances anti-tumor immunity by blocking IL-6/STAT3-driven immunosuppression. Additionally, combining kinase inhibitors targeting the PI3K/AKT/mTOR pathway with ICIs shows significant synergy in TNBC by modulating the tumor microenvironment to sensitize cancer cells to immunotherapy, thereby enhancing immune response and inhibiting tumor proliferation [140, 146]. Wnt and Notch inhibitors when paired with PI3K or HER2-targeted therapies, reduce CSC populations, improving outcomes in resistant subtypes. Biomarker-driven approaches such as profiling PIK3CA, RAS or STAT3 alterations are essential for patient selection to ensure these combinations are effective [7, 15, 147].

Beyond pathway crosstalk, additional synergistic strategies enhance BC treatment. Natural compounds like quercetin and cannabidiol exhibit synergistic effects with chemotherapeutic agents like doxorubicin in MCF-7 cells, reducing toxicity and enhancing antiproliferative effects by targeting oxidative stress and apoptosis pathways [148, 149]. Nanoparticle-based delivery systems such as ellagic acid-loaded chitosan nanoparticles or platelet-superparamagnetic iron oxide nanoparticles improve targeted delivery of chemotherapeutics like paclitaxel, enabling chemo-hyperthermia therapy that enhances tumor cell death while minimizing systemic toxicity [150, 151]. In TNBC, combining PARP inhibitors with ICIs overcomes BRCA-mutated resistance by inducing DNA damage that boosts neoantigen presentation, enhancing immunotherapy efficacy [152]. Photodynamic therapy, paired with sustained delivery of metformin and catalase-like activity via G-quadruplex hydrogels, generates reactive oxygen species and modulates metabolic pathways for tumor suppression [153]. Hyperthermia, when combined with chemotherapy or immunotherapy, disrupts tumor cell membranes and enhances drug penetration, showing promise in preclinical models [154]. Drug repurposing strategies such as combining antineoplastic drugs with central nervous system drugs, demonstrate synergistic growth inhibition in MCF-7 and HT-29 cells by leveraging complementary mechanisms [155]. Engineered microalgae for photo-sonodynamic therapy combine light and ultrasound to induce tumor cell death, offering a novel approach [156]. Next-generation inhibitors such as PROTACs targeting PI3K, AKT or β -catenin provide precise protein degradation to overcome resistance from mutations [79]. RNA-based therapies including siRNAs targeting STAT3 or Notch1 show promise in silencing oncogenic drivers [157]. Clinical trials are exploring combinations like ATR inhibitors with PARP inhibitors to exploit synthetic lethality in DNA repair-deficient tumors, addressing resistance across pathways. Nanoparticle-based systems enhance these combinations by improving drug bioavailability and tumor-specific delivery particularly for TNBC [158]. Multi-omics and AI-driven analyses advance personalized strategies by identifying pathway interactions and predictive biomarkers. Despite challenges like optimizing dosing schedules and ensuring clinical translatability, these synergistic approaches which address resistance mechanisms such as EGFR upregulation or multidrug resistance proteins by targeting multiple pathways simultaneously, hold immense potential to

transform BC therapy by personalizing treatment and improving progression-free survival while reducing adverse effects.

7 Acquired resistance factors with mechanistic solutions in breast cancer

Acquired resistance to therapeutic interventions in BC poses a significant challenge to achieving sustained clinical responses particularly in HR+, HER2+ and TNBC subtypes. The miscellaneous nature of resistance encompasses genetic mutations, aberrant signaling pathways, TME dynamics, epigenetic reprogramming and metabolic adaptations. These mechanisms enable cancer cells to evade endocrine therapies, targeted agents, chemotherapy and immunotherapies. Categorizing these resistance factors and elucidating their molecular underpinnings are critical for developing innovative therapeutic strategies. This discussion provides an in-depth exploration of these categories, proposing mechanistic solutions informed by recent research:

7.1 Genetic mutations as drivers of resistance

Genetic mutations are pivotal in acquired resistance particularly in HR + and HER2 + BC. Mutations in the *ESR1* gene encoding the ER are prevalent in HR + metastatic BC and confer resistance to endocrine therapies such as aromatase inhibitors and SERMs. These mutations, often in the ligand-binding domain (e.g., Y537S and D538G) result in ligand-independent activation, rendering therapies like tamoxifen and fulvestrant ineffective [159]. Similarly, *PIK3CA* mutations activate the PI3K/AKT/mTOR pathway, driving resistance to both endocrine and HER2-targeted therapies such as trastuzumab [160]. In TNBC, mutations in tumor protein 53 or heterozygous retinoblastoma 1 loss impair DNA repair and cell cycle regulation, contributing to resistance against chemotherapy and PARP inhibitors [161]. Additionally, BRCA1/2 mutations or reversion mutations in DNA repair genes further complicate PARP inhibitor efficacy in TNBC [162]. The MDR1 gene when overexpressed, enhances drug efflux, reducing chemotherapy efficacy [163]. Targeting *ESR1* mutations necessitates next-generation SERDs like camizestran which effectively degrade mutant receptors and inhibit downstream signaling, as evidenced in clinical trials [164]. For *PIK3CA*-driven resistance, combining PI3K inhibitors (e.g., alpelisib) with CDK4/6 inhibitors restores sensitivity in HR + BC by targeting compensatory signalling [165]. Novel PROTACs targeting PARP1 induce ferroptosis in p53-positive cancers, offering a strategy to overcome PARP inhibitor resistance by promoting programmed cell death [166]. Gene-editing technologies such as CRISPR hold potential for correcting resistance-conferring mutations, though clinical translation remains nascent [167]. PROTACs targeting MDR1 degrade efflux pumps, enhancing chemotherapy efficacy [163]. These precision medicine approaches highlight the need to tailor therapies to specific genetic alterations.

7.2 Aberrant signaling pathway activation

Aberrant activation of signaling pathways enables cancer cells to bypass therapeutic inhibition, a hallmark of acquired resistance. The PI3K/AKT/mTOR and MAPK/MEK/ERK pathways are frequently hyperactivated, compensating for the blockade of primary therapeutic targets. In HER2 + BC, HER2-AXL heterodimerization activates alternative survival pathways, reducing the efficacy of HER2 inhibitors like trastuzumab [168]. In HR + BC, resistance to CDK4/6 inhibitors (e.g., palbociclib) is driven by retinoblastoma

1 loss or high p16 expression, disrupting cell cycle checkpoints [169]. The JAK-STAT pathway, particularly in inflammatory BC, promotes chemotherapy-resistant cell states by enhancing survival and stemness [170]. Upregulation of FGFR4 through N6-methyladenosine modifications attenuates ferroptotic cell death, contributing to resistance in HER2 + BC [171]. Additionally, heat shock protein beta-1 facilitates chemoresistance by inhibiting ferroptosis and regulating NF- κ B signaling [172]. Dual inhibition of complementary pathways is a promising strategy. Combining mTOR inhibitors with BRAF/MEK inhibitors resensitizes BRAF-mutant cancers, a strategy applicable to TNBC with similar pathway dependencies [173]. Inhibition of CK1 ϵ enhances CDK4/6 inhibitor efficacy by restoring cell cycle regulation in HR + BC [174]. Targeting AXL with small-molecule inhibitors disrupts HER2-AXL heterodimerization, restoring sensitivity to HER2-directed therapies [168]. JAK inhibitors suppress chemotherapy-resistant states in inflammatory BC by blocking JAK-STAT signaling [170]. N6-methyladenosine demethylase inhibitors such as those targeting fat mass and obesity-associated protein, destabilize FGFR4, promoting ferroptosis in HER2 + BC [171]. These combination therapies emphasize the importance of targeting multiple signaling nodes to prevent compensatory activation.

7.3 Tumour microenvironment and immune evasion

The TME significantly contributes to resistance by fostering a protective niche for cancer cells. TAMs secrete cytokines (e.g., IL-6, TGF- β) that promote survival and drug resistance in BC cells [175]. In TNBC, the TME drives immune checkpoint blockade resistance by upregulating PD-L1 expression and creating an immunosuppressive environment, limiting the efficacy of anti-PD-1/PD-L1 therapies [176]. CSCs within the TME, characterized by their quiescent and drug-resistant states, evade chemotherapy and targeted therapies [177]. CAFs enhance resistance by remodelling the extracellular matrix, reducing drug penetration [178]. Additionally, stromal components contribute to metabolic reprogramming, supporting tumor cell survival under therapeutic stress [179]. Modulating the TME offers innovative approaches to overcome resistance. Combining anti-PD-1/PD-L1 antibodies with MAPK inhibitors enhances immune checkpoint blockade efficacy in TNBC by restoring immune activation [165]. Fasting-mimicking diets reduce CSC populations in TNBC, enhancing chemotherapy sensitivity by disrupting their protective niche [177]. Colony-stimulating factor 1 receptor inhibitors target TAMs, reducing their pro-tumorigenic activity and restoring sensitivity to therapies [175]. Matrix metalloproteinase inhibitors disrupt stromal remodeling, improving drug penetration and efficacy [178]. Targeting metabolic pathways such as glucose metabolism, with glycolysis inhibitors disrupts the TME's support for resistant cells [180]. These TME-targeted interventions combined with existing therapies hold significant promise.

7.4 Epigenetic and metabolic reprogramming

Epigenetic modifications including DNA methylation, histone modifications and non-coding RNA regulation dynamically contribute to resistance. Long non-coding RNAs such as *EILA* stabilize cyclin E1 protein, promoting CDK4/6 inhibitor resistance in HR + BC [180]. N6-methyladenosine modifications regulate FGFR4 expression, attenuating ferroptotic cell death in HER2 + BC [171]. HDAC-mediated epigenetic silencing

of tumor suppressor genes such as CDKN1A/p21 contributes to endocrine resistance [181]. Metabolic reprogramming particularly glucose metabolic shifts supports resistance by enhancing energy production and biosynthetic pathways [179]. Overexpression of MDR1 via epigenetic regulation enhances drug efflux, reducing chemotherapy efficacy [163]. Fat mass and obesity-associated protein inhibitors targeting N6-methyladenosine demethylation destabilize resistance-associated proteins, restoring sensitivity to targeted therapies [182]. PROTACs targeting MDR1 degrade efflux pumps, enhancing chemotherapy efficacy [163]. HDAC inhibitors reverse epigenetic silencing of tumor suppressor genes, resensitizing cells to endocrine therapies [183]. Glycolysis inhibitors disrupt metabolic reprogramming, enhancing chemotherapy efficacy by limiting energy supply to resistant cells [179]. Additionally, targeting long non-coding RNAs like EILA with RNA interference or small molecules can restore CDK4/6 inhibitor sensitivity [178]. These approaches highlight the potential of epigenetic and metabolic interventions in overcoming resistance.

7.5 Intratumoral heterogeneity

Intratumoral heterogeneity, both genetic and phenotypic is a critical driver of acquired resistance in BC. Tumors comprise diverse subpopulations of cells with varying genetic profiles, gene expression patterns and drug sensitivities, allowing some clones to survive therapeutic pressure. In HER2 + BC heterogeneity in HER2 expression enables certain cell populations to evade anti-HER2 therapies like trastuzumab by switching to alternative driver pathways such as EGFR or MET signalling [184]. In HR + BC, clonal evolution under endocrine therapy pressure selects for resistant subclones harbouring ESR1 mutations or PIK3CA alterations [159]. In TNBC, heterogeneity in tumor epithelial and microenvironmental components contributes to chemotherapy resistance as resistant clones exploit diverse survival mechanisms [185]. Targeting intratumoral heterogeneity requires strategies that address multiple subclones simultaneously. Combination therapies such as dual HER2 blockade with trastuzumab and pertuzumab reduce the likelihood of resistant clone emergence in HER2 + BC [184]. In HR + BC, integrating SERDs like camizestrant with PI3K inhibitors can target both ESR1-mutant and PIK3CA-mutant subclones [164]. Liquid biopsies and single-cell sequencing enable real-time monitoring of clonal dynamics, guiding adaptive treatment strategies [167]. Additionally, therapies targeting non-genetic heterogeneity, such as Bromodomain and extra-terminal inhibitors that disrupt transcriptional plasticity show promise in reducing resistant subpopulations [185]. These approaches aim to suppress the evolutionary adaptability of heterogeneous tumors.

7.6 Non-genetic plasticity and reversible resistance

Non-genetic plasticity refers to the ability of cancer cells to adopt reversible adaptive states that confer resistance without permanent genetic changes. In HR + BC, non-genetic heterogeneity driven by epigenetic and transcriptional reprogramming allows cells to enter drug-tolerant persister states which are reversible upon drug withdrawal [186]. These persister cells, often characterized by stem-like properties, contribute to resistance against CDK4/6 inhibitors and endocrine therapies [169]. In TNBC, plasticity enables cancer cells to switch between epithelial and mesenchymal states, promoting resistance to chemotherapy and immune checkpoint inhibitors [176]. This plasticity is

mediated by dynamic changes in gene expression, often regulated by epigenetic modifiers or microRNAs [187]. Targeting non-genetic plasticity involves disrupting the adaptive mechanisms that sustain drug-tolerant persister states. Inhibiting epigenetic regulators such as EZH2 or HDACs can prevent the transition to resistant states in HR + BC [183]. Combining CDK4/6 inhibitors with Bromodomain and extra-terminal inhibitors disrupts transcriptional reprogramming, reducing persisted cell survival [186]. In TNBC, targeting EMT with TGF- β inhibitors resensitizes cells to chemotherapy by limiting plasticity [176]. MicroRNA-based therapies such as those targeting miR-200 family members can suppress EMT and enhance sensitivity to immune checkpoint blockade [187]. These strategies aim to lock cancer cells in drug-sensitive states, preventing reversible resistance.

7.7 Drug efflux mechanisms

Drug efflux pumps, beyond MDR1, contribute to resistance by expelling chemotherapeutic agents and targeted therapies from cancer cells. In addition to MDR1, other ATP-binding cassette transporters such as ABCG2 (BC resistance protein) and MRP7 are upregulated in resistant BC cells, reducing intracellular drug concentrations [182]. In TNBC, ABCG2 overexpression is a key mechanism of resistance to anthracyclines and taxanes [161]. In HER2 + BC, efflux pumps diminish the efficacy of ADCs like T-DM1 by expelling cytotoxic payloads [162]. Inhibiting efflux pumps is a viable strategy to restore drug sensitivity. Meclofenamic acid, which downregulates ABCG2 and MRP7 via at mass and obesity-associated protein/N6-methyladenosine-demethylation has shown promise in resensitizing cells to chemotherapy [179]. PROTACs targeting MDR1 and other ABC transporters degrade efflux pumps, enhancing intracellular drug accumulation [163]. Nanoparticle-based drug delivery systems can bypass efflux mechanisms by improving drug uptake and retention in cancer cells [188]. These approaches aim to maximize therapeutic efficacy by ensuring adequate drug exposure.

7.8 Autophagy as a survival mechanism

Autophagy, a cellular process for degrading and recycling damaged components promotes resistance by enabling cancer cells to survive therapeutic stress. In HR + BC, resistance to endocrine therapies is associated with increased autophagy which mitigates cellular stress induced by estrogen deprivation [183]. In TNBC, autophagy protects cancer cells from chemotherapy-induced apoptosis particularly in hypoxic TME regions [177]. Autophagy also contributes to resistance against CDK4/6 inhibitors by maintaining cellular homeostasis in HR + BC [180]. Chloroquine and hydroxychloroquine autophagy inhibitors sensitize TNBC cells to chemotherapy by promoting apoptosis [177]. In HR + BC, combining autophagy inhibitors with endocrine therapies or CDK4/6 inhibitors disrupts resistant cell survival [183]. Targeting upstream regulators of autophagy such as mTOR or AMPK with specific inhibitors (e.g., everolimus) can further enhance therapeutic outcomes [173]. These strategies exploit autophagy's role as a survival mechanism to resensitize resistant cells.

7.9 Stromal interactions beyond immune TME

Beyond immune-related TME components, stromal interactions involving ECM remodelling and stromal cell signalling contribute to resistance. CAFs secrete growth factors

such as HGF and FGF that activate survival pathways in cancer cells, promoting resistance to targeted therapies [178]. ECM stiffening, driven by CAFs and matrix metalloproteinases reduces drug penetration and enhances resistance to chemotherapy and ADCs [162]. In metastatic BC, bone marrow stromal cells protect cancer cells from endocrine and chemotherapy-induced stress via cytokine signalling [178]. Matrix metalloproteinase inhibitors improve drug penetration by reducing ECM stiffness, enhancing chemotherapy and ADC efficacy [162]. Inhibiting CAF-derived growth factors such as HGF with MET inhibitors resensitizes cells to HER2-targeted therapies [184]. In metastatic cancer, bisphosphonates or RANKL inhibitors disrupt bone marrow stromal support, enhancing endocrine therapy efficacy [178]. These strategies target the non-immune stromal components to overcome resistance.

7.10 Tumor cell dormancy

Tumor cell dormancy is a critical mechanism of acquired resistance, particularly in HR+ (Luminal A and B) and metastatic BC. Dormant tumor cells enter a quiescent state, characterized by low proliferation (low Ki-67 expression) and resistance to therapies targeting actively dividing cells such as chemotherapy and CDK4/6 inhibitors [159]. In Luminal A and B tumours which rely on ER signalling driven by ESR1, GATA3 and FOXA1 dormant cells can persist in minimal residual disease states, particularly in the bone marrow, surviving endocrine therapies like SERMs or aromatase inhibitors [160, 178]. These cells often exhibit stem-like properties and can reactivate years later, contributing to late recurrences, a hallmark of HR + BC with 5-year survival rates exceeding 90% for Luminal A but lower for Luminal B due to higher proliferation [159]. In TNBC, dormancy allows cells to evade chemotherapy-induced apoptosis particularly in basal-like subtypes with tumour protein 53 mutations (~ 80%) and high genomic instability [161, 169]. In HER2-enriched tumors, driven by ERBB2 amplification and PI3K/AKT/mTOR or RAS/MAPK activation, dormant cells resist anti-HER2 therapies like trastuzumab by entering a low-proliferative state, contributing to metastatic relapse [164, 184]. Targeting dormant cells requires disrupting their survival mechanisms or forcing re-entry into the cell cycle. In HR + BC, combining endocrine therapies with inhibitors of survival pathways such as BCL2 inhibitors (e.g., venetoclax) can induce apoptosis in dormant cells by targeting anti-apoptotic proteins upregulated in Luminal subtypes [167]. In TNBC, epigenetic therapies such as HDAC inhibitors can reactivate dormant cells sensitizing them to chemotherapy [183]. For HER2-enriched tumors, dual HER2 blockade with trastuzumab and pertuzumab combined with mTOR inhibitors prevents dormant cell survival by blocking PI3K/AKT signalling [173]. Additionally, integrin inhibitors targeting $\alpha\beta3$ which mediate dormant cell adhesion in bone marrow niches show promise in preventing reactivation [178]. These strategies aim to eradicate quiescent populations to reduce recurrence risk.

7.11 Cancer stem cell (CSC) dynamics

CSCs are a subset of tumor cells with self-renewal and differentiation capabilities, driving resistance across all BC subtypes. In Luminal A and B tumors, CSCs contribute to endocrine resistance by maintaining a quiescent, drug-tolerant state, often mediated by upregulation of stemness markers like ALDH1 and CD44/CD24-low [186]. These CSCs resist SERMs and AIs due to reduced dependence on ER signaling and enhanced

survival pathways such as Notch or Wnt, activated via crosstalk with PIK3CA mutations (40% in HR + tumours) [160, 167]. In HER2-enriched tumours, CSCs overexpress ERBB2 and resist anti-HER2 therapies by activating alternative pathways such as HER2-AXL heterodimerization [168]. In TNBC, particularly basal-like subtypes, CSCs drive chemotherapy resistance due to high expression of EMT genes (SNAIL, SLUG) and DNA repair defects from BRCA1/2 mutations (20%) [161, 169]. CSCs also contribute to TNBC's metastatic potential by promoting angiogenesis (VEGF) [169]. Targeting CSCs involves disrupting their self-renewal and survival. In HR + BC, Notch inhibitors (e.g., gamma-secretase inhibitors) reduce CSC populations, enhancing endocrine therapy efficacy [167]. In HER2-enriched tumors, AXL inhibitors combined with trastuzumab disrupt CSC-driven resistance by blocking alternative survival pathways [168]. In TNBC, fasting-mimicking diets reduce CSC populations by altering the TME, sensitizing cells to chemotherapy [177]. Additionally, targeting BRCA1/2-mutant CSCs with PARP inhibitors such as Olaparib exploits DNA repair defects, though resistance via reversion mutations requires novel approaches like PARP1 or PROTACs to induce ferroptosis [166]. These strategies aim to eliminate CSCs to prevent resistance and metastasis.

7.12 Exosomal signalling

Exosomal signalling involving the release of extracellular vesicles (exosomes) carrying bioactive molecules (e.g., MicroRNAs, proteins) mediates resistance by facilitating intercellular communication. In Luminal A and B tumours, exosomes transfer microRNAs such as miR-221/222 that downregulate PTEN, enhancing PI3K/AKT/mTOR signaling and endocrine resistance [187]. In HER2-enriched tumors, exosomes carrying ERBB2 mRNA or HER2 protein promote resistance to anti-HER2 therapies by transferring oncogenic signals to neighbouring cells [184]. In TNBC, exosomes enriched with SNAIL or VEGF mRNA enhance EMT and angiogenesis, contributing to chemotherapy resistance and metastasis [169]. Exosomes also mediate TME remodeling by transferring drug efflux pumps like MDR1 to sensitive cells, reducing chemotherapy efficacy [163]. Inhibiting exosomal signaling is a novel approach to overcome resistance. In HR + BC, exosome biogenesis inhibitors such as GW4869 reduce microRNA transfer, restoring PTEN expression and endocrine sensitivity [187]. In HER2-enriched tumors, targeting exosomal ERBB2 with RNA interference or exosome uptake inhibitors (e.g., annexin V) enhances trastuzumab efficacy [184]. In TNBC, neutralizing exosomal VEGF with anti-VEGF antibodies (e.g., bevacizumab) reduces angiogenesis and chemotherapy resistance [169]. Additionally, nanoparticle-based therapies that disrupt exosome release or uptake can enhance drug retention by limiting MDR1 transfer [163]. These approaches target exosomal communication to restore therapeutic sensitivity.

7.13 Tumor-immune interactions beyond immunosuppression

Tumor-immune interactions, beyond TME-mediated immunosuppression contribute to resistance through adaptive immune evasion and immune cell dysfunction. In Luminal B tumours which often co-express HER2 or EGFR, low TILs and upregulation of immune checkpoints (PD-L1, Cytotoxic T-Lymphocyte-Associated Protein 4) reduce immunotherapy efficacy [160, 167]. In HER2-enriched tumors, immune evasion occurs via downregulation of major histocompatibility complex class I molecules, limiting T-cell recognition despite high ERBB2 expression [164]. In TNBC, while basal-like subtypes

show high TILs, dysfunctional T-cells and Treg infiltration impair immune checkpoint blockade (e.g., anti-PD-1/PD-L1) efficacy [166, 169]. Additionally, tumour-derived cytokines such as TGF- β reprogram immune cells to a pro-tumorigenic phenotype, enhancing resistance across subtypes [11]. Enhancing tumor-immune interactions improves therapeutic outcomes. In Luminal B tumors, combining anti-PD-L1 antibodies with PI3K inhibitors boosts TIL infiltration and immunotherapy response [165]. In HER2-enriched tumors, oncolytic viruses or major histocompatibility complex class I agonists restore T-cell recognition, enhancing anti-HER2 therapy efficacy [164]. In TNBC, combining anti-PD-1 antibodies with MAPK inhibitors overcomes T-cell dysfunction, improving immune checkpoint blockade [165]. TGF- β inhibitors such as galunisertib reprogram the immune microenvironment, reducing Treg infiltration and enhancing chemotherapy sensitivity in TNBC [169]. These strategies leverage immune activation to overcome resistance.

8 Immune modulation and the tumor microenvironment

The TME is a dynamic ecosystem comprising immune cells, fibroblasts, endothelial cells and ECM components that interact with tumor cells to influence BC progression, immune evasion and therapeutic resistance. Immune modulation within the TME is a critical determinant of tumor behavior, particularly in immunogenic subtypes like TNBC. This section explores the roles of TILs, TAMs, CAFs and ECM-driven angiogenesis along with their therapeutic implications.

8.1 Tumor-infiltrating lymphocytes (TILs)

TILs including cytotoxic CD8 + T cells, CD4 + helper T cells and Tregs play a dual role in BC. CD8 + T cells recognize and eliminate tumor cells via antigen-specific cytotoxicity while CD4 + T cells support immune responses through cytokine production. However, Tregs suppress anti-tumor immunity, promoting immune evasion. In TNBC, high TIL infiltration is associated with better prognosis and response to immunotherapy but tumors often upregulate immune checkpoints like PD-L1 to evade detection [8]. ICIs such as pembrolizumab and atezolizumab target the PD-1/PD-L1 axis to restore T-cell activity and enhance anti-tumor immunity. These agents have shown significant efficacy in metastatic TNBC particularly in patients with high tumor mutational burden and PD-L1 expression. Combining ICIs with chemotherapy or radiation therapy enhances immunogenic cell death, further boosting immune responses [189, 190].

8.2 Tumor-associated macrophages (TAMs)

TAMs particularly M2-polarized macrophages promote immunosuppression, angiogenesis and tumor progression by secreting anti-inflammatory cytokines (e.g., IL-10) and growth factors (e.g., VEGF). M2 TAMs suppress cytotoxic T-cell activity and support tumor cell survival through ECM remodeling and immune checkpoint upregulation. In BC, high M2 TAM infiltration is associated with poor prognosis and resistance to therapy [191]. Colony-stimulating factor 1 receptor inhibitors target TAM recruitment and polarization, reducing immunosuppression and enhancing T-cell activity. These inhibitors are being explored in combination with ICIs to reprogram the TME and overcome resistance particularly in TNBC [192].

8.3 Cancer-associated fibroblasts (CAFs)

CAFs remodel the ECM, secrete cytokines (e.g., TGF- β) and promote tumor growth, invasion and metastasis. They contribute to TME stiffness which enhances tumor cell migration and immune suppression. CAFs also interact with tumor cells and immune cells to create an immunosuppressive niche, limiting T-cell infiltration [193]. CXCR4 inhibitors target CAF-derived chemokine signaling, disrupting their tumor-supportive functions. These inhibitors are under investigation to enhance immunotherapy efficacy by reducing TME-mediated immunosuppression [194].

8.4 Extracellular matrix and angiogenesis

The ECM provides structural support and modulates tumor behavior through stiffness and remodeling. Vascular endothelial growth factor-driven angiogenesis supports tumor growth by supplying oxygen and nutrients. ECM stiffness promotes metastasis by facilitating tumor cell invasion and immune evasion. Anti-VEGF therapies such as Bevacizumab inhibit angiogenesis, reducing tumor vascularization and growth. These therapies are being combined with ICIs to enhance immune infiltration by normalizing tumor vasculature. The TME's role in immune modulation highlights its importance as a therapeutic target. By targeting TAMs, CAFs and angiogenesis, therapies can reprogram the TME to enhance anti-tumor immunity, linking immune modulation to metabolic and epigenetic mechanisms [196].

9 Epigenetic modifications in breast cancer

Epigenetic modifications including DNA methylation, histone modifications and non-coding RNA regulation play a critical role in BC progression, immune evasion and therapeutic resistance. DNA methylation involves the addition of methyl groups to CpG islands in gene promoters, leading to transcriptional silencing. In BC, hypermethylation of tumor suppressor gene promoters (e.g., BRCA1, RASSF1A) inhibits their expression, promoting tumor progression and resistance to therapy. Hypomethylation of oncogene promoters can also drive tumorigenesis [197, 198]. DNA methyltransferase inhibitors such as 5-azacytidine reverse promoter hypermethylation, reactivating tumor suppressor genes. These inhibitors are being explored in combination with other therapies to enhance treatment efficacy, particularly in TNBC where epigenetic silencing is prevalent [197]. Histone modifications including acetylation and deacetylation regulate chromatin structure and gene expression. HDACs remove acetyl groups from histones, leading to chromatin condensation and transcriptional repression of tumor suppressor genes. In BC, HDAC-mediated silencing of genes like BIM promotes tumor survival and resistance to therapy [198, 199]. HDACs, particularly class I HDACs repress tumor suppressor genes like BIM in TNBC by removing acetyl groups from histones, leading to chromatin condensation and transcriptional silencing. This epigenetic modification inhibits the expression of genes critical for apoptosis and cell cycle regulation, promoting tumor survival and resistance to therapy. HDAC inhibitors such as Entinostat and Vorinostat increase histone acetylation, reactivating tumor suppressor genes and promoting apoptosis. In TNBC, Entinostat has shown promise by upregulating genes involved in apoptosis (e.g., BIM) and enhancing immune-related gene expression, sensitizing tumors to ICIs [198, 200]. Entinostat block HDAC activity, increasing histone acetylation and promoting an open chromatin state. This reactivates tumor suppressor

genes such as BIM, a pro-apoptotic protein, leading to increased tumor cell death. Additionally, Entinostat enhances the expression of major histocompatibility complex class I molecules and tumor-associated antigens, making TNBC cells more recognizable to cytotoxic CD8 + T cells [1, 2, 5]. Entinostat enhances the efficacy of ICIs such as anti-PD-1 (e.g., Nivolumab) and anti-cytotoxic T-lymphocyte-associated protein 4 (e.g., Ipilimumab) antibodies by reprogramming the TME to favor anti-tumor immunity. It reduces the activity of immunosuppressive cells including Tregs and myeloid-derived suppressor cells by downregulating immunosuppressive genes like Foxp3. Entinostat also upregulates PD-L1 expression on tumor cells, increasing their susceptibility to PD-1/PD-L1 inhibitors. Furthermore, it promotes CD8 + T cell infiltration and activity by enhancing antigen presentation and reducing TME immunosuppression. The synergy is driven by Entinostat's effects on both tumor cells and the TME including reducing lactate-mediated acidification (linked to the Warburg effect) which otherwise suppresses T-cell function and modulating cytokine signaling (e.g., IL-6/JAK/STAT3) to enhance T-cell activation [1, 5]. A phase Ib clinical trial demonstrated that Entinostat combined with Nivolumab and Ipilimumab improved response rates in women with advanced HER2-negative BC including TNBC by enhancing T-cell-mediated tumor clearance. This combination overcomes the immunosuppressive TME which limits the efficacy of immunotherapy alone in TNBC by enhancing tumor immunogenicity and reducing resistance mechanisms. The combination of Entinostat with ICIs offers a promising strategy for TNBC where limited targeted therapies exist. By reactivating tumor suppressors and reprogramming the TME, Entinostat sensitizes TNBC to immune-mediated destruction, improving patient outcomes. Ongoing clinical trials are optimizing dosing, reducing toxicity and identifying biomarkers for patient selection [5]. Non-coding RNAs including microRNAs and long non-coding RNAs regulate gene expression at the transcriptional and post-transcriptional levels. In BC, microRNAs like miR-148a target oncogenes (e.g., BCL-2) while long non-coding RNAs like LINC00511 promote tumor progression through epigenetic regulation [201]. RNA-based therapies such as small interfering RNAs and antisense oligonucleotides target oncogenic microRNAs and long non-coding RNAs to restore tumor suppressor function or inhibit oncogene expression. These therapies are being developed for TNBC to overcome resistance and enhance treatment efficacy [92]. Epigenetic modifications mediate tumor-TME interactions and resistance mechanisms, making them critical therapeutic targets. By reversing epigenetic silencing, therapies can restore anti-tumor immunity and sensitize tumors to other treatments, bridging to advanced therapeutic strategies.

10 Metabolic reprogramming in breast cancer

Metabolic reprogramming is a hallmark of cancer, enabling tumor cells to meet the bio-energetic and biosynthetic demands of rapid proliferation, survival, and adaptation to the TME. In BC, metabolic alterations are intricately linked to oncogenic signaling pathways and contribute to tumor progression, therapeutic resistance and immune evasion. This section explores the key metabolic pathways, their subtype-specific profiles and their therapeutic implications with a focus on glucose, amino acid and lipid metabolism as well as the interplay between tumor cells and the TME.

10.1 Glucose metabolism (Warburg effect)

BC cells exhibit a preference for aerobic glycolysis, a phenomenon known as the Warburg effect where glucose is metabolized to lactate even in the presence of oxygen. This process, driven by upregulation of glucose transporter 1, hexokinase 2 and lactate dehydrogenase A results in increased lactate production and acidification of the TME. This acidic environment promotes immune evasion, angiogenesis and metastasis by altering the ECM and suppressing immune cell function. Hypoxia-inducible factor 1 alpha further enhances glycolytic gene expression under hypoxic conditions, supporting tumor growth by providing metabolic intermediates for biosynthesis such as citrate for lipid synthesis and ATP for cellular processes [202, 203]. Targeting glycolysis offers a promising therapeutic strategy. Lactate dehydrogenase A inhibitors such as FX11 disrupt lactate production, impairing tumor cell survival and TME acidification. Similarly, glycolytic inhibitors like 2-deoxy-D-glucose competitively inhibit glucose metabolism, reducing energy availability for cancer cells. These inhibitors are under investigation for their potential to sensitize tumors to other therapies, particularly in aggressive subtypes like TNBC where glycolysis is predominant [201].

10.2 Amino acid metabolism

BC cells, particularly TNBC exhibit a strong dependency on glutamine metabolism to support rapid proliferation and survival. Glutamine serves as a nitrogen and carbon source for nucleotide and amino acid synthesis. Enzymes like glutaminase, which converts glutamine to glutamate are overexpressed in TNBC, driving biosynthetic pathways. Additionally, serine and glycine metabolism, facilitated by phosphoglycerate dehydrogenase amplification, supports nucleotide synthesis in ER- tumors. Arginine metabolism is also altered in argininosuccinate synthetase 1-deficient tumors which rely on exogenous arginine for survival. Glutaminase inhibitors such as CB-839 (Telaglenastat) are being evaluated in clinical trials to block glutamine metabolism and induce cancer cell death particularly in TNBC. Arginine-degrading enzymes such as Polyethylene glycolylated arginine deiminase exploit arginine dependency in ASS1-deficient tumors, leading to metabolic stress and apoptosis. These therapies aim to disrupt the metabolic dependencies of cancer cells, offering a targeted approach to overcome resistance [204].

10.3 Lipid metabolism

Lipid metabolism is significantly altered in BC to support membrane synthesis, signaling and energy demands. Fatty acid synthase and acetyl-CoA carboxylase are overexpressed, promoting de novo fatty acid synthesis. CD36, a fatty acid transporter facilitates lipid uptake while lipophagy (lipid droplet breakdown) provides energy through lipid catabolism. These alterations support tumor growth, particularly in HER2 + BC where lipogenesis is driven by PI3K/AKT/mTOR signalling. Fatty acid synthase inhibitors such as TVB-2640 and ACC inhibitors are under investigation to disrupt lipid synthesis and storage, impairing tumor growth. Combining Fatty acid synthase inhibitors with ICIs is a promising strategy to enhance anti-tumor immunity by altering lipid-dependent signaling pathways. These approaches are particularly relevant for HER2 + and TNBC subtypes where lipid metabolism is a critical driver of tumor progression [205].

10.4 Subtype-specific metabolic profiles

Metabolic reprogramming in BC is subtype-specific, reflecting the molecular heterogeneity of the disease. TNBC exhibits a highly glycolytic phenotype with reliance on glutaminolysis, driven by oncogenic signaling pathways like MYC and PI3K/AKT. HER2 + tumors show enhanced lipogenesis, supported by PI3K/AKT/mTOR activation to meet the demands of rapid proliferation. In contrast, ER + tumors predominantly utilize oxidative phosphorylation and fatty acid oxidation, reflecting their slower proliferation rates and reliance on mitochondrial metabolism. These distinct metabolic profiles influence therapeutic responses and resistance mechanisms [206].

10.5 TME and metabolic crosstalk

The TME plays a critical role in metabolic reprogramming through bidirectional interactions between tumor cells and stromal components. The “Reverse Warburg” effect describes how CAFs undergo aerobic glycolysis to produce lactate and ketones which are then utilized by tumor cells for oxidative phosphorylation. Exosomes, small extracellular vesicles further mediate metabolic crosstalk by transferring enzymes like pyruvate kinase M2 which enhance glycolysis in recipient cells. Lactate secretion by tumor cells lowers the TME pH, promoting immunosuppression and invasion. Additionally, short-chain fatty acids from the gut microbiota modulate tumor metabolism by acting as HDAC inhibitors, altering metabolic gene expression. The interplay between metabolic reprogramming and the TME underscores the importance of targeting metabolic pathways to disrupt tumor survival and enhance immunotherapy efficacy. By altering the metabolic landscape of the TME, therapies can sensitize tumors to immune-mediated destruction, linking metabolic alterations to immune modulation and resistance mechanisms [207, 208].

11 Spatial transcriptomics insights into immune cell subsets and treatment response biomarkers

Spatial transcriptomics has revolutionized BC research by preserving spatial context in gene expression profiles, enabling a detailed understanding of the TME and its role in treatment response and resistance. Unlike single-cell RNA sequencing which loses spatial information, ST maps transcriptomes to specific tissue locations, revealing cellular interactions and heterogeneity critical for identifying biomarkers. This analysis focuses on Tregs and M2 macrophages, key immunosuppressive immune cell subsets and their association with treatment response biomarkers in BC, drawing on recent ST studies to highlight their spatial distribution, functional roles and therapeutic implications.

11.1 Tregs in breast cancer TME

Tregs, characterized by high FOXP3 expression are critical immunosuppressive cells in the BC TME, promoting tumor immune evasion by suppressing cytotoxic T cell activity [133]. Spatial transcriptomics studies reveal their spatial organization within BC tissues particularly in TNBC where they are enriched in stromal and peritumoral regions, often co-localizing with TAMs and CAFs. Huang et al. (2023) identified Tregs with enhanced tumor spatial specificity in TNBC, suggesting direct induction by neoplastic cells. Their proximity to tumor boundaries correlates with immunosuppressive niches, driven by IL-10 and TGF- β signalling which inhibit CD8 + T cell function and promote

tumour progression [133, 209]. In HER2 + BC, Andersson et al. (2021) reported Treg interactions with macrophages during type I interferon responses, indicating a spatially restricted immunosuppressive role [210]. Tregs are associated with poor immunotherapy response in TNBC particularly to ICIs like anti-PD-1/PD-L1 therapies. Mebane et al. (2025) used ST to analyze TNBC tumors with tertiary lymphoid structures, finding that Treg enrichment in stromal regions correlates with reduced ICI efficacy due to PD-L1-mediated immunosuppression [209]. High Treg infiltration predicts worse clinical outcomes as they suppress anti-tumor immunity, even in tertiary lymphoid structure-rich tumours typically associated with better prognosis [133, 209]. Huang et al. (2023) identified Treg-specific gene signatures including FOXP3 and cytotoxic T-lymphocyte-associated protein 4 as biomarkers of immunotherapy resistance, with spatial data showing their accumulation near tumour margins where they form immunosuppressive barriers [133]. In HR + BC, Tregs contribute to endocrine therapy resistance by enhancing CAF-mediated stromal remodelling as shown by Mao et al. (2023) in metastatic axillary lymph nodes where Treg-CAF interactions upregulate ECM-related genes [210]. Targeting Tregs is a promising strategy to enhance immunotherapy response. Spatial transcriptomics-guided approaches suggest combining anti-cytotoxic T-lymphocyte-associated protein 4 antibodies with ICIs to deplete Tregs and restore CD8 + T cell activity [133, 209]. Mao et al. (2023) proposed combining Treg-targeted therapies with PI3K inhibitors as Treg-CAF crosstalk in metastatic BC upregulates PI3K/AKT signaling, contributing to resistance [210]. Biomarker-driven trials using ST to quantify Treg spatial distribution and FOXP3 expression could optimize patient selection for combination therapies particularly in TNBC and metastatic settings [209, 211].

11.2 M2 macrophages in breast cancer TME

M2 macrophages characterized by CD163, CD206 and anti-inflammatory markers (e.g., IL-10, ARG1) promote tumor growth, angiogenesis and immune suppression in BC [212, 213]. ST studies reveal their heterogeneous distribution with enrichment in peritumoral and stromal regions particularly in TNBC and HER2 + BC [210, 211, 213]. Matusiak et al. (2023) identified M2-like macrophage niches in BC, expressing MRC1 and CCL18, located near tumour boundaries where they engage in “don’t-eat-me” signaling via CD47-SIRPA interactions with tumour cells [213]. Chu et al. (2024) highlighted the spatiotemporal heterogeneity of M2 macrophages, noting their accumulation in invasive zones and association with lipid metabolism (e.g., FABP5 expression) which supports tumour cell survival [212]. In metastatic BC, Mao et al. (2023) found M2 macrophages co-localized with CAFs in axillary lymph nodes, forming tumour immune barriers that enhance metastasis through secreted phosphoprotein 1 (Osteopontin) signalling [210]. Wang et al. (2024) reported that PD-L1-expressing M2 macrophages in TNBC are associated with better clinical outcomes when treated with ICIs as their spatial proximity to tertiary lymphoid structure enhances immune activation [214]. However, high M2 macrophage infiltration generally correlates with poor prognosis due to their immunosuppressive role [212, 213]. Verona et al. (2025) identified M2 macrophage-CSC interactions mediated by Notch4 and type I interferon signaling as drivers of chemoresistance in BC, with ST revealing their co-localization in tumour niches [215]. Gao et al. (2025) used spatial multi-omics to show that M2 macrophages in BC lung metastases upregulate lipid metabolism genes, predicting resistance to platinum-based therapies [216].

ST-derived biomarkers such as CD163, CCL18 and FABP5 are associated with treatment failure, guiding the development of macrophage-targeted therapies [212, 213]. Reprogramming M2 macrophages to an M1-like pro-inflammatory phenotype is a key strategy to overcome resistance. Chu et al. (2024) proposed targeting colony-stimulating factor 1 receptor to inhibit M2 macrophage recruitment, with ST data supporting its efficacy in reducing tumour-promoting niches [212]. Combining colony-stimulating factor 1 receptor inhibitors with ICIs or PI3K/AKT inhibitors, as suggested by Mao et al. (2023) could disrupt M2 macrophage-CAF interactions in metastatic BC [210]. Verona et al. (2025) investigated for Notch4 inhibitors to target M2 macrophage-CSC crosstalk, enhancing chemotherapy sensitivity [215]. Spatial transcriptomics-guided spatial profiling of CD163 and secreted phosphoprotein 1 expression can identify patients likely to benefit from these combination therapies, particularly in TNBC and metastatic subtypes [210, 212].

11.3 Pathway crosstalk and resistance mechanisms

ST studies highlight synergistic interactions between Tregs and M2 macrophages in promoting immunosuppression. Zhang et al. (2024) noted that Tregs and M2 macrophages co-localize in TNBC stromal regions, forming immunosuppressive niches driven by IL-10 and TGF- β signalling. These interactions upregulate PD-L1 expression, reducing ICI efficacy [209, 217, 218]. Mao et al. (2023) identified tumor immune barrier structures in metastatic BC where Tregs and M2 macrophages interact with CAFs via secreted phosphoprotein 1 and insulin-like growth factor 1 receptor signalling, enhancing PI3K/AKT pathway activation and resistance to targeted therapies [210]. Verona et al. (2025) further showed that Notch4-mediated crosstalk between M2 macrophages and CSCs sustains Treg infiltration, promoting a feedback loop that drives chemoresistance [215]. The PI3K/AKT/mTOR pathway, frequently dysregulated in BC, interacts with Treg and M2 macrophage signaling to drive resistance. Mao et al. (2023) found that M2 macrophage-derived cytokines activate PI3K/AKT in tumor cells, enhancing CSC survival and resistance to Alpelisib in HR + BC [210]. Huang et al. (2023) reported that Treg-mediated TGF- β signaling upregulates PI3K/AKT in TNBC, correlating with ICI resistance [133]. ST data from Gao et al. (2025) revealed that M2 macrophage lipid metabolism genes (e.g., FABP5) activate PI3K/AKT, promoting tumor cell survival in metastatic niches [216]. Combining PI3K inhibitors with Treg or M2 macrophage-targeted therapies could disrupt these interactions, improving treatment outcomes [210, 217].

12 Role of AI in breast cancer therapy

12.1 AI in breast cancer diagnosis

AI, particularly machine learning and deep learning, processes multimodal data—genomic, transcriptomic, proteomic, and imaging—to identify predictive biomarkers and optimize therapeutic strategies. For instance, AI models predict responses to therapies like CDK4/6 inhibitors, chemotherapy (e.g., Paclitaxel), and endocrine therapies by analyzing molecular signatures such as HER2, ER and tumor-infiltrating lymphocytes, achieving high accuracy (e.g., AUC of 0.85–0.90 for treatment response prediction). In the neoadjuvant setting, AI integrates clinical and imaging data to forecast pathologic complete response with studies reporting up to 88% accuracy for Paclitaxel-based regimens, enabling tailored treatment plans that minimize toxicity and improve outcomes. AI also enhances immunotherapy

personalization by identifying biomarkers like tumor mutational burden and PD-L1 expression, guiding immune checkpoint inhibitor use in TNBC [219]. Furthermore, AI-driven radiomics extracts quantitative features from mammograms, ultrasounds, and MRIs, improving risk stratification and treatment selection. Machine learning and deep learning models analyze genomic data to identify biomarkers like HER2, BRCA1/2, and PIK3CA mutations, enabling prediction of responses to targeted therapies such as trastuzumab or PARP inhibitors with accuracies up to 90% (e.g., AUC 0.90 for trastuzumab response). In immunotherapy, AI integrates tumor mutational burden and PD-L1 expression to guide immune checkpoint inhibitor use, particularly for TNBC, improving response rates by 15–20% compared to traditional methods. Radiomics, powered by convolutional neural networks extracts quantitative features from imaging (e.g., MRI, mammography), achieving 85–92% accuracy in predicting neoadjuvant chemotherapy outcomes, such as pathologic complete response to anthracycline-based regimens. AI also personalizes endocrine therapy by analyzing hormone receptor status (ER/PR) and clinical variables, optimizing treatment duration and reducing recurrence risk [220].

12.2 AI in biomarker discovery

AI excels at identifying tissue-based biomarkers such as cellular morphology and protein expression patterns, critical for early breast cancer detection and staging. Using deep learning models like convolutional neural networks, AI analyzes histopathological images and imaging data (e.g., mammograms) to detect subtle patterns often missed by human clinicians. For example, AI-driven tools like PathAI outperform human pathologists in analyzing tissue samples, spotting smaller or less obvious malignancies with higher accuracy, leading to earlier intervention and improved outcomes. AI-derived biomarkers help predict disease progression and recurrence risk, particularly for aggressive subtypes like TNBC. By analyzing molecular and imaging data, AI forecasts outcomes, enabling tailored treatment plans to reduce the likelihood of cancer returning or worsening. AI integrates diverse datasets such as imaging, genomics, and proteomics to create comprehensive biomarker profiles. For instance, combining HER2 expression with histopathological features improves the classification of breast cancer subtypes, enhancing diagnostic precision and guiding targeted therapies. Breast cancer's tumor heterogeneity demands extensive data analysis to uncover actionable biomarkers. AI, particularly deep learning, processes high-dimensional omics data (e.g., gene expression profiles, proteomic data) and large-scale datasets from next-generation sequencing. This reveals complex patterns that traditional statistical methods struggle to detect, accelerating the discovery of novel biomarkers for personalized treatment [221]. Moreover, AI-driven platforms, like Tempus, combine tumor genomic profiles with clinical outcomes to identify biomarkers that guide personalized treatment plans, improving response rates in clinical trials. Jyothi NM notes that AI's analysis of multi-omics data provides deeper insights into tumor biology, enabling the discovery of meta-biomarkers that capture complex interactions across biological pathways, crucial for addressing breast cancer's heterogeneity. AI's has potential role in pharmacogenomics where algorithms predict patient-specific responses to drugs based on genomic biomarkers, optimizing therapies like anti-HER2 antibodies for better progression-free survival and overall survival [222]. AI improves the accuracy of histopathological analysis by automating the detection of biomarkers such as HER2, estrogen receptor ER, PR and Ki67, which are critical for breast cancer diagnosis and treatment planning. The paper cites Wu et al. (2024), who evaluated AI

algorithms (e.g., HER2-CONNECT, CNN, HER2 4B5) across 6,867 cases, demonstrating increased accuracy and reproducibility in HER2 detection compared to manual immunohistochemistry methods [4]. AI models, such as those from Lunit Inc. (e.g., Lunit SCOPE PD-L1 and Lunit SCOPE IO), analyze whole slide images (WSIs) to quantify programmed cell death-ligand 1 expression and tumor-infiltrating lymphocyte density, providing prognostic insights and reducing interobserver variability. These models use colored dots and patches to highlight cell types and segmented areas, enhancing pathologists' concordance rates [223].

12.3 AI in ADC design and development & predicting response

AI can integrate multi-omics data with clinical variables (e.g., patient demographics, tumor stage, prior treatments) to identify biomarkers predictive of ADC efficacy. For instance, AI models can correlate antigen expression levels or TME characteristics with response rates, enabling clinicians to select patients most likely to benefit from specific ADCs. AI-driven tools can provide real-time predictions during clinical decision-making. For example, by analyzing imaging data (e.g., PET scans) or liquid biopsies, AI can dynamically assess tumor characteristics and recommend whether an ADC is suitable for a patient. AI is revolutionizing the design and optimization of ADCs by accelerating the identification of optimal monoclonal antibodies by predicting binding affinity to tumor antigens. Computational tools like AlphaFold or molecular dynamics simulations can model antibody-antigen interactions, reducing the need for costly and time-consuming experimental screening. AI can predict linker stability and cleavage kinetics based on chemical properties, while also suggesting novel payloads with enhanced potency or reduced off-target effects. AI can play a pivotal role in understanding and overcoming resistance mechanisms, such as reduced antigen expression, altered drug internalization, or efflux pump activation. AI can analyze multi-omics data to uncover molecular pathways driving resistance. For example, deep learning models can identify upregulated efflux pumps (e.g., P-glycoprotein) or mutations in apoptosis pathways that reduce ADC efficacy [224].

12.4 AI in overcoming drug resistance

Machine learning models identify mutations in genes like ESR1 or PIK3CA that drive resistance to endocrine therapies (e.g., tamoxifen) or targeted agents (e.g., trastuzumab). Deep learning algorithms, such as convolutional neural networks and recurrent neural networks can detect epigenetic changes such as DNA methylation patterns that contribute to chemotherapy resistance in TNBC. AI plays a great role in analyzing single-cell RNA sequencing data to identify resistant subpopulations within breast tumors. By pinpointing cells with upregulated efflux pumps or alternative signaling pathways, AI provides insights into resistance mechanisms, enabling targeted interventions. AI models such as those using deep reinforcement learning or generative adversarial networks predict how breast cancer cells respond to specific drugs. For example, AI tools like PERCEPTION use single-cell transcriptomics to predict resistance to therapies in breast cancer, outperforming traditional bulk sequencing by identifying distinct tumor cell subpopulations. AI integrates clinical, imaging and molecular data to predict resistance risk. For instance, combining MRI radiomic features with HER2 status and Ki67 index achieves high predictive accuracy (AUC of 0.986 in training, 0.935 in validation) for chemotherapy resistance in breast cancer. AI accelerates drug discovery by identifying existing drugs that can be repurposed to target resistant breast cancer cells. AI helps in analyzing multi-omics data to design novel compounds such

as Z29077885 which targets STK33 to promote apoptosis and halt cell cycle progression in resistant tumors. AI algorithms including variational autoencoders and generative adversarial networks facilitate de novo drug design by modeling drug-target interactions, optimizing molecules to bypass resistance mechanisms like efflux pumps. AI predicts synergistic drug combinations to overcome resistance. For example, by modeling molecular pathways, AI suggests combining endocrine therapies with PI3K inhibitors to block alternative survival pathways in hormone receptor-positive breast cancer. AI-driven platforms analyze patient-specific tumor biology to recommend combination therapies, reducing the likelihood of resistance. AI collaborates with nanotechnology to design intelligent nanocarriers that enhance drug delivery to resistant cells. For instance, AI optimizes nanoparticle compositions to target specific tumor features such as overexpressed TROP2 in TNBC, improving drug uptake and reducing systemic toxicity. AI assists in tailoring cisplatin-loaded solid lipid nanoparticles to overcome resistance while minimizing side effects. AI identifies biomarkers like TROP2-QCS which predicts response to ADCs like datopotamab deruxtecan in breast cancer. Patients with high TROP2-QCS expression showed a 43% reduction in disease progression risk compared to 25% in those without, demonstrating AI's ability to stratify patients for targeted therapies. AI analyzes proteomic and genomic data to identify biomarkers like PD-L1 or TIL density that predict resistance to immunotherapy or chemotherapy. These biomarkers guide treatment adjustments such as switching to alternative therapies for resistant TNBC. AI platforms like Tempus and Caris use state-of-the-art algorithms to mine patient molecular data for resistance biomarkers, informing clinical trial design and personalized treatment plans [225].

12.5 AI in nanomedicine discovery

AI designs nanocarriers such as liposomes and quantum dots that serve dual diagnostic and therapeutic roles (theranostics). These nanocarriers enable real-time monitoring of tumor response through advanced imaging, guided by AI's predictive modeling to optimize diagnostic accuracy. AI-powered theranostic nanocarriers, such as quantum dots and upconversion nanoparticles, enable simultaneous diagnosis and therapy, improving treatment precision and monitoring. Machine learning algorithms including artificial neural networks optimize nanoparticle properties such as size, shape and surface chemistry to improve drug delivery to breast cancer cells. For example, AI tailors liposomes and gold nanoparticles to target overexpressed receptors like TROP2 in TNBC, enhancing drug uptake and reducing off-target effects. FDA-approved nanomedicines like Doxil® and Abraxane® demonstrate improved pharmacokinetics and reduced toxicity, further enhanced by AI-driven design. AI designs stimuli-activatable nanocarriers that respond to tumor-specific cues (e.g., low pH, high redox levels) or external stimuli (e.g., ultrasound, light). AI optimizes polymeric micelles to release drugs like doxorubicin precisely at tumor sites, improving therapeutic efficacy and minimizing systemic toxicity. AI predicts drug-nanoparticle interactions, ensuring optimal drug loading and release kinetics. For instance, machine-learning based design of experiments optimizes nano-formulations, adhering to Quality by Design principles to define dosage and administration parameters, reducing development timelines and costs. AI-powered nanocarriers provide real-time feedback on treatment response. For instance, theranostic nanoparticles enable simultaneous drug delivery and imaging, allowing AI to monitor drug distribution and adjust dosing dynamically. AI optimizes nanocarriers to bypass resistance mechanisms such as efflux pumps or altered signaling pathways. For example, AI designs

cisplatin-loaded solid lipid nanoparticles to enhance drug uptake in resistant TNBC cells, reducing systemic toxicity. Furthermore, AI streamlines the nanomedicine development pipeline through machine learning-based design of experiments, reducing costs and time-lines. For instance, optimization algorithms automate the formulation process, ensuring high-quality nanocarriers with minimal failures [226].

13 Conclusion

BC comprises a spectrum of molecularly defined subtypes with distinct biological drivers, therapeutic sensitivities and resistance mechanisms. The delineation of Luminal A, Luminal B, HER2-enriched and TNBC has enabled subtype-specific treatment strategies, yet long-term efficacy is often limited by intrinsic and acquired resistance. These resistance mechanisms are frequently mediated by genomic alterations (e.g., ESR1, PIK3CA, BRCA1/2 mutations), compensatory activation of parallel signaling pathways (e.g., PI3K/AKT/mTOR, RAS/MAPK, Notch) and interactions with components of the TME including immune cells and stromal factors. This review integrates mechanistic insights into key oncogenic pathways and their roles in promoting uncontrolled proliferation, survival, metastasis and therapeutic resistance. It also outlines the clinical relevance of targeting these pathways with advanced modalities including next-generation SERDs, ADCs, kinase inhibitors and immune checkpoint inhibitors. Combinatorial strategies that concurrently inhibit multiple nodes within and across signaling pathways have demonstrated enhanced efficacy in preclinical models and clinical trials, particularly in hormone receptor-positive and HER2-positive subtypes. Furthermore, incorporation of molecular profiling and predictive biomarkers is essential for selecting appropriate targeted therapies and identifying patients at risk of resistance. Integration of multi-omics data and machine learning algorithms is advancing precision oncology by enabling real-time therapeutic adaptation based on tumor evolution. In conclusion, optimizing BC management requires a mechanistic understanding of subtype-specific biology, resistance networks and treatment response patterns. Future therapeutic development should focus on rational drug combinations and continued evaluation of emerging agents through adaptive clinical trial frameworks to improve progression-free and overall survival outcomes.

Abbreviations

ABC	ATP-Binding Cassette
ADC	Antibody-drug conjugate
AI	Aromatase inhibitor
BC	Breast cancer
BCL2	B-cell lymphoma 2
CAF	Cancer-associated fibroblast
CCND1	Cyclin D1
CDK	Cyclin-dependent kinase
CSC	Cancer stem cell
ECM	Extracellular matrix
EGFR	Epidermal growth factor receptor
EMT	Epithelial-mesenchymal transition
ER	Estrogen receptor
ESR1	Estrogen receptor 1
FGFR	Fibroblast growth factor receptor
HDAC	Histone deacetylase
HER2	Human epidermal growth factor receptor 2
HR	Hormone receptor
ICI	Immune checkpoint inhibitor
IL	Interleukin
JAK	Janus kinase
MAPK	Mitogen-activated protein kinase
MDR1	Multidrug resistance protein 1

MEK	Mitogen-activated protein kinase kinase
MET	Mesenchymal-epithelial transition (receptor)
mTOR	Mammalian target of rapamycin
MYC	Myelocytomatosis oncogene
NF	κ B-Nuclear Factor Kappa B
ORR	Objective response rate
PD	1-Programmed Death-1
PD	L1-Programmed Death-Ligand 1
PI3K	Phosphoinositide 3-Kinase
PR	Progesterone receptor
PROTAC	Proteolysis-targeting chimera
PTEN	Phosphatase and tensin homolog
RANKL	Receptor Activator of Nuclear Factor Kappa-B Ligand
RAS	Rat sarcoma (oncogene)
SERD	Selective estrogen receptor degrader
SERM	Selective estrogen receptor modulator
siRNA	Small interfering RNA
STAT	Signal transducer and activator of transcription
TAM	Tumor-associated macrophage
T	DM1-trastuzumab emtansine
T	DXd-trastuzumab deruxtecan
TGF	β -transforming growth factor beta
TIL	Tumor-infiltrating lymphocyte
TME	Tumor microenvironment
TNBC	Triple-negative breast cancer
Treg	Regulatory T cell
VEGF	Vascular endothelial growth factor

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Declarations

Ethics approval and consent to participate

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All the authors have given their consent for publication.

Competing interests

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References

1. Harbeck N, Penault-Llorca F, Cortes J, Gnant M, Houssami N, Poortmans P, et al. Breast cancer. *Nat Rev Dis Primers*. 2019;5(1):66. <https://doi.org/10.1038/s41572-019-0111-2>.
2. Turner NC, Reis-Filho JS. Genetic heterogeneity and cancer drug resistance. *Lancet Oncol*. 2012;13(4):e178–85. doi: 10.1016/S1470-2045(11)70335-7. PMID: 22469128.
3. Herzog SK, Fuqua SA. ESR1 mutations and therapeutic resistance in metastatic breast cancer: progress and remaining challenges. *Br J Cancer*. 2022;126(2):174–86. <https://doi.org/10.1038/s41416-021-01564-x>. PMID: 34621045.
4. Rios-Hoyo A, Shan NL, Karn PL, Pusztai L. Clinical implications of breast cancer intrinsic subtypes. In: Pusztai L, editor. *A guide to breast cancer research: from cellular heterogeneity and molecular mechanisms to therapy*. 1st ed. Cham: Springer; 2025. pp. 435–48.
5. Busonero C, Leone S, Bartoloni S, Acconcia F. Strategies to degrade estrogen receptor α in primary and ESR1 mutant-expressing metastatic breast cancer. *Mol Cell Endocrinol*. 2019;480:107–21. <https://doi.org/10.1016/j.mce.2018.10.020>. PMID: 30389467.

6. Saatci O, Huynh-Dam KT, Sahin O. Endocrine resistance in breast cancer: from molecular mechanisms to therapeutic strategies. *J Mol Med (Berl)*. 2021;99(12):1691–710. <https://doi.org/10.1007/s00109-021-02136-5>. PMID: 34623477.
7. Wang M, Li J, Huang J, Luo M. The predictive role of PIK3CA mutation status on PI3K inhibitors in HR+ breast cancer therapy: a systematic review and meta-analysis. *Biomed Res Int*. 2020;2020:1598037. <https://doi.org/10.1155/2020/1598037>. PMID: 32461963.
8. Guarneri V, Griguolo G, Miglietta F, Conte PF, Dieci MV, Girardi F. Survival after neoadjuvant therapy with trastuzumab–lapatinib and chemotherapy in patients with HER2-positive early breast cancer: A meta-analysis of randomized trials. *ESMO open*. 2022;7(2). <https://doi.org/10.1016/j.esmoop.2022.100433>. PMID: 35276440.
9. Tarantino P, Gandini S, Nicolo E, Trillo P, Giugliano F, Zagami P et al. Evolution of low HER2 expression between early and advanced-stage breast cancer. *Eur J Cancer*. 2022;163:35–43. doi: 10.1016/j.ejca.2021.12.022. PMID: 35032815.
10. Guo L, Kong D, Liu J et al. Breast cancer heterogeneity and its implication in personalized precision therapy. *Exp Hematol Oncol*. 2023;12(1):3. <https://doi.org/10.1186/s40164-022-00363-1>. PMID: 36624542.
11. Gaynor N, Crown J, Collins DM. Immune checkpoint inhibitors: key trials and an emerging role in breast cancer. *Semin Cancer Biol*. 2022;79:44–57. <https://doi.org/10.1016/j.semcancer.2020.06.016>. PMID: 32623044.
12. Mavratzas A, Marme F. Alpelisib in the treatment of metastatic HR+ breast cancer with PIK3CA mutations. *Future Oncol*. 2021;17(1):13–36. <https://doi.org/10.2217/fon-2020-0464>. PMID: 32964734.
13. Saini KS, Loi S, de Azambuja E et al. Targeting the PI3K/AKT/mTOR and Raf/MEK/ERK pathways in the treatment of breast cancer. *Cancer Treat Rev*. 2013;39(8):935–46. doi: 10.1016/j.ctrv.2013.03.009. PMID: 23643661.
14. Zhang X, Dong N, Hu X. Wnt/β-catenin signaling inhibitors. *Curr Top Med Chem*. 2023;23(10):880–96. <https://doi.org/10.2174/1568026623666230303101810>. PMID: 36872364.
15. Shao F, Pang X, Baeg GH. Targeting the JAK/STAT signaling pathway for breast cancer. *Curr Med Chem*. 2021;28(25):5137–51. <https://doi.org/10.2174/0929867328666201207202012>. PMID: 33290193.
16. Coussy F, El Botty R, Lavigne M, Gu C, Fuhrmann L, Briaux A et al. Combination of PI3K and MEK inhibitors yields durable remission in PDX models of PIK3CA-mutated metaplastic breast cancers. *J Hematol Oncol*. 2020;13(1):13. <https://doi.org/10.1186/s13045-020-0846-y>. PMID: 32087759.
17. Mao X, Zhou D, Lin K et al. Single-cell and spatial transcriptomics analyses revealed cell heterogeneity and immune environment alternations in metastatic axillary lymph nodes in breast cancer. *Cancer Immunol Immunother*. 2023;72(3):679–95. <https://doi.org/10.1007/s00262-022-03278-2>. PMID: 36040519.
18. Alum EU. AI-driven biomarker discovery: enhancing precision in cancer diagnosis and prognosis. *Discov Oncol*. 2025;16(1):1–2. <https://doi.org/10.1007/s12672-025-02064-7>. PMID: 40082367.
19. Mehmood S, Faheem M, Ismail H, Farhat SM, Ali M, Younis S, Asghar MN. Breast cancer resistance likelihood and personalized treatment through integrated multiomics. *Front Mol Biosci*. 2022;9:783494. <https://doi.org/10.3389/fmolb.2022.783494>. PMID: 35495618.
20. Bravo MJ, Burgos-Molina AM, García-Aranda M et al. Unlocking new avenues in breast cancer treatment: the synergy of kinase inhibitors and immunotherapy. *Cancers (Basel)*. 2023;15(23):5499. <https://doi.org/10.3390/cancers15235499>. PMID: 38067203.
21. Deng Z, Chen G, Shi Y et al. Curcumin and its nano-formulations: defining triple-negative breast cancer targets through network pharmacology, molecular docking, and experimental verification. *Front Pharmacol*. 2022;13:920514. <https://doi.org/10.3389/fphar.2022.920514>. PMID: 36003508.
22. Perou CM, Sørlie T, van de Eisen MB, Jeffrey SS, Rees CA et al. Molecular portraits of human breast tumours. *Nature*. 2000;406(6797):747–52. <https://doi.org/10.1038/35021093>. PMID: 10963602.
23. Sørlie T, Perou CM, Tibshirani R, Aas T, Geisler S, Johnsen H et al. Gene expression patterns of breast carcinomas distinguish tumor subclasses with clinical implications. *Proc Natl Acad Sci U S A*. 2001;98(19):10869–74. <https://doi.org/10.1073/pnas.191367098>. PMID: 11553815.
24. Hammond ME, Hayes DF, Dowsett M, Allred DC, Hagerty KL, Badve S et al. American Society of Clinical Oncology/College of American Pathologists guideline recommendations for immunohistochemical testing of estrogen and progesterone receptors in breast cancer. *J Clin Oncol*. 2010;28(16):2784–95. <https://doi.org/10.1200/JCO.2009.25.6529>. PMID: 20404251.
25. Vazhappilly CG, Saleh E, Ramadan W, Menon V, Al-Azawi AM, Tarazi H et al. Inhibition of SHP2 by new compounds induces differential effects on RAS/RAF/ERK and PI3K/AKT pathways in different cancer cell types. *Invest New Drugs*. 2019;37:252–61. <https://doi.org/10.1007/s10637-018-0626-5>. PMID: 29947013.
26. Miranda F, Prazeres H, Mendes F, Martins D, Schmitt F. Resistance to endocrine therapy in HR+ and/or HER2+ breast cancer: the most promising predictive biomarkers. *Mol Biol Rep*. 2022;49(1):717–33. <https://doi.org/10.1007/s11033-021-06863-3>. PMID: 34739691.
27. Yang X, Weng X, Yang Y, Zhang M, Xiu Y, Peng W, Liao X, Xu M, Sun Y, Liu X. A combined hypoxia and immune gene signature for predicting survival and risk stratification in triple-negative breast cancer. *Aging (Albany NY)*. 2021;13(15):19486. <https://doi.org/10.18632/aging.203360>. PMID: 34341184.
28. Zheng S, Zou Y, Liang JY, Xiao W, Yang A, Meng T et al. Identification and validation of a combined hypoxia and immune index for triple-negative breast cancer. *Mol Oncol*. 2020;14(11):2814–33. <https://doi.org/10.1002/1878-0261.12747>. PMID: 32521117.
29. Clusan L, Ferrière F, Flouriot G, Pakdel F. A basic review on estrogen receptor signaling pathways in breast cancer. *Int J Mol Sci*. 2023;24(7):6834. <https://doi.org/10.3390/ijms24076834>. PMID: 37047814.
30. Dimitrakopoulos FI, Kottorou A, Tzeou A. Endocrine resistance and epigenetic reprogramming in estrogen receptor positive breast cancer. *Cancer Lett*. 2021;517:55–65. <https://doi.org/10.1016/j.canlet.2021.05.030>. PMID: 34077785.
31. Hanker AB, Sudhan DR, Arteaga CL. Overcoming Endocrine Resistance in Breast Cancer. *Cancer Cell*. 2020;37(4):496–513. <https://doi.org/10.1016/j.ccell.2020.03.009>. PMID: 32289273.
32. Khatpe AS, Adebayo AK, Herodotou CA, Kumar B, Nakshatri H. Nexus between PI3K/AKT and estrogen receptor signaling in breast cancer. *Cancers (Basel)*. 2021;13(3):369. <https://doi.org/10.3390/cancers13030369>. PMID: 33514082.
33. Khan MZ, Uzair M, Nazli A, Chen JZ. An overview on Estrogen receptors signaling and its ligands in breast cancer. *Eur J Med Chem*. 2022;241:114658. <https://doi.org/10.1016/j.ejmech.2022.114658>. PMID: 35964426.
34. Chakravorty G, Ahmad S, Godbole MS, Gupta S, Badwe RA, Dutt A. Deciphering the mechanisms of action of progesterone in breast cancer. *Oncotarget*. 2023;14:660–667. <https://doi.org/10.18632/oncotarget.28455>. PMID: 37395734.

35. Keklikoglou I, Koerner C, Schmidt C, Zhang JD, Heckmann D, Shavinskaya A et al. MicroRNA-520/373 family functions as a tumor suppressor in estrogen receptor negative breast cancer by targeting NF- κ B and TGF- β signaling pathways. *Oncogene*. 2012;31(37):4150–63. <https://doi.org/10.1038/ncr.2011.571>. PMID: 22158050.
36. Li Z, Wei H, Li S, Wu P, Mao X. The Role of Progesterone Receptors in Breast Cancer. *Drug Des Devel Ther*. 2022;16:305–314. <https://doi.org/10.2147/DDDT.S336643>. PMID: 35115765.
37. Grimm SL, Hartig SM, Edwards DP. Progesterone Receptor Signaling Mechanisms. *J Mol Biol*. 2016;428(19):3831–49. <https://doi.org/10.1016/j.jmb.2016.06.020>. PMID: 27380738.
38. Salazar M, Lerma-Ortiz A, Hooks GM, Ashley AK, Ashley RL. Progesterone-mediated activation of MAPK and AKT in nuclear progesterone receptor negative breast epithelial cells: The role of membrane progesterone receptors. *Gene*. 2016;591(1):6–13. <https://doi.org/10.1016/j.gene.2016.06.044>. PMID: 27349565.
39. Ordaz-Ramos A, Diaz-Blancas J, Martínez-Cruz A, Castro-Oropeza R, Zampieri C, Romero-Rodríguez DP et al. RANKL regulates differentially breast cancer stem cell properties through its RANK and LGR4 receptors. *Biochim Biophys Acta Mol Cell Res*. 2025;1872(2):119888. <https://doi.org/10.1016/j.bbamcr.2024.119888>. PMID: 39662745.
40. Hernández-Hernández OT, Camacho-Arroyo I. Regulation of gene expression by progesterone in cancer cells: effects on cyclin D1, EGFR and VEGF. *Mini Rev Med Chem*. 2013;13(5):635–42. <https://doi.org/10.2174/1389557511313050002>. PMID: 23432380.
41. Cenciari ME, Proietti CJ. Molecular mechanisms underlying progesterone receptor action in breast cancer: Insights into cell proliferation and stem cell regulation. *Steroids*. 2019;152:108503. <https://doi.org/10.1016/j.steroids.2019.108503>. PMID: 31562879.
42. Miziak P, Baran M, Błaszczak E, Przybyszewska-Podstawka A, Kałafut J, Smok-Kalwat J et al. Estrogen Receptor Signaling in Breast Cancer. *Cancers (Basel)*. 2023;15(19):4689. <https://doi.org/10.3390/cancers15194689>. PMID: 37835383.
43. Yarden Y, Slivkowski MX. Untangling the ErbB signalling network. *Nat Rev Mol Cell Biol*. 2001;2(2):127–37. doi: 10.1038/35052073. PMID: 11252954.
44. Moasser MM. The oncogene HER2: its signaling and transforming functions and its role in human cancer pathogenesis. *Oncogene*. 2007;26(45):6469–87. <https://doi.org/10.1038/sj.onc.1210477>. PMID: 17471238.
45. Iqbal N, Iqbal N. Human Epidermal Growth Factor Receptor 2 (HER2) in Cancers: Overexpression and Therapeutic Implications. *Mol Biol Int*. 2014;2014:852748. <https://doi.org/10.1155/2014/852748>. PMID: 25276427.
46. Pan L, Li J, Xu Q, Gao Z, Yang M, Wu X, Li X. HER2/PI3K/AKT pathway in HER2-positive breast cancer: A review. *Medicine (Baltimore)*. 2024;103(24):e38508. <https://doi.org/10.1097/MD.00000000000038508>. PMID: 38875362.
47. Slamon DJ, Clark GM, Wong SG, Levin WJ, Ullrich A, McGuire WL. Human breast cancer: correlation of relapse and survival with amplification of the HER-2/neu oncogene. *Science*. 1987;235(4785):177–82. <https://doi.org/10.1126/science.3798106>. PMID: 3798106.
48. Cheng X. A Comprehensive Review of HER2 in Cancer Biology and Therapeutics. *Genes (Basel)*. 2024;15(7):903. <https://doi.org/10.3390/genes15070903>. PMID: 39062682.
49. Sergina NV, Moasser MM. The HER family and cancer: emerging molecular mechanisms and therapeutic targets. *Trends Mol Med*. 2007;13(12):527–34. <https://doi.org/10.1016/j.molmed.2007.10.002>. PMID: 17981505.
50. Lee JJ, Loh K, Yap YS. PI3K/Akt/mTOR inhibitors in breast cancer. *Cancer Biol Med*. 2015;12(4):342–54. <https://doi.org/10.7497/j.issn.2095-3941.2015.0089>. PMID: 26779371.
51. Fillbrunn M, Signorovitch J, André F, Wang I, Lorenzo I, Ridolfi A et al. PIK3CA mutation status, progression and survival in advanced HR + /HER2- breast cancer: a meta-analysis of published clinical trials. *BMC Cancer*. 2022;22(1):1002. <https://doi.org/10.1186/s12885-022-10078-5>. PMID: 36131248.
52. Chien TJ. A review of the endocrine resistance in hormone-positive breast cancer. *Am J Cancer Res*. 2021;11(8):3813–31. PMID: 34522451.
53. Alves CL, Ditzel HJ. Drugging the PI3K/AKT/mTOR Pathway in ER+ Breast Cancer. *Int J Mol Sci*. 2023;24(5):4522. <https://doi.org/10.3390/ijms24054522>. PMID: 36901954.
54. Li Q, Li Z, Luo T, Shi H. Targeting the PI3K/AKT/mTOR and RAF/MEK/ERK pathways for cancer therapy. *Mol Biomed*. 2022;3(1):47. <https://doi.org/10.1186/s43556-022-00110-2>. PMID: 36539659.
55. Rocca A, Braga L, Volpe MC, Maiocchi S, Generali D. The Predictive and Prognostic Role of RAS-RAF-MEK-ERK Pathway Alterations in Breast Cancer: Revision of the Literature and Comparison with the Analysis of Cancer Genomic Datasets. *Cancers (Basel)*. 2022;14(21):5306. <https://doi.org/10.3390/cancers14215306>. PMID: 36358725.
56. Giltman JM, Balko JM. Rationale for targeting the Ras/MAPK pathway in triple-negative breast cancer. *Discov Med*. 2014;17(95):275–83. PMID: 24882719.
57. Leung EY, Kim JE, Askarian-Amiri M, Rewcastle GW, Finlay GJ, Baguley BC. Relationships between signaling pathway usage and sensitivity to a pathway inhibitor: examination of trametinib responses in cultured breast cancer lines. *PLoS One*. 2014;9(8):e105792. <https://doi.org/10.1371/journal.pone.0105792>. PMID: 25170609.
58. Raut D, Vora A, Bhatt LK. The Wnt/ β -catenin pathway in breast cancer therapy: a pre-clinical perspective of its targeting for clinical translation. *Expert Rev Anticancer Ther*. 2022;22(1):97–114.
59. Arzi L, Mollaei H, Hoshyar R. Countering Triple Negative Breast Cancer via Impeding Wnt/ β -Catenin Signaling, a Phytotherapeutic Approach. *Plants (Basel)*. 2022;11(17):2191. <https://doi.org/10.3390/plants11172191>. PMID: 36079579.
60. Zhang Y, Wang X. Targeting the Wnt/ β -catenin signaling pathway in cancer. *J Hematol Oncol*. 2020;13(1):165. <https://doi.org/10.1186/s13045-020-00990-3>. PMID: 33276800.
61. Xing Z, Wang X, Liu J, Liu G, Zhang M, Feng K et al. Effect of MiR-210 on the Chemosensitivity of Breast Cancer by Regulating JAK-STAT Signaling Pathway. *Biomed Res Int*. 2021;2021:7703159. <https://doi.org/10.1155/2021/7703159>. PMID: 34423038.
62. López-Mejía JA, Mantilla-Ollarves JC, Rocha-Zavaleta L. Modulation of JAK-STAT Signaling by LNK: A Forgotten Oncogenic Pathway in Hormone Receptor-Positive Breast Cancer. *Int J Mol Sci*. 2023;24(19):14777. <https://doi.org/10.3390/ijms241914777>. PMID: 37834225.
63. Dinakar YH, Kumar H, Mudavath SL, Jain R, Ajmeer R, Jain V. Role of STAT3 in the initiation, progression, proliferation and metastasis of breast cancer and strategies to deliver JAK and STAT3 inhibitors. *Life Sci*. 2022;309:120996. <https://doi.org/10.1016/j.lfs.2022.120996>. PMID: 36170890.

64. Tabassum S, Abbasi R, Ahmad N, Farooqi AA. Targeting of JAK-STAT Signaling in Breast Cancer: Therapeutic Strategies to Overcome Drug Resistance. *Adv Exp Med Biol*. 2019;1152:271–281. https://doi.org/10.1007/978-3-030-20301-6_14. PMID: 31456189.
65. Edwards A, Brennan K. Notch Signalling in Breast Development and Cancer. *Front Cell Dev Biol*. 2021;9:692173. <https://doi.org/10.3389/fcell.2021.692173>. PMID: 34295896.
66. Nandi A, Chakrabarti R. The many facets of Notch signaling in breast cancer: toward overcoming therapeutic resistance. *Genes Dev*. 2020;34(21–22):1422–1438. <https://doi.org/10.1101/gad.342287.120>. PMID: 33872192.
67. Kontomanolis EN, Kalagasidou S, Pouliliou S, Anthoulaki X, Georgiou N, Papamanolis V, Fasoulakis ZN. The Notch Pathway in Breast Cancer Progression. *ScientificWorldJournal*. 2018;2018:2415489. <https://doi.org/10.1155/2018/2415489>. PMID: 30111989.
68. Mayer EL. Targeting breast cancer with CDK inhibitors. *Curr Oncol Rep*. 2015;17(5):443. <https://doi.org/10.1007/s11912-015-0443-3>. PMID: 25716100.
69. Izadi S, Nikkhoo A, Hojjat-Farsangi M, Namdar A, Azizi G, Mohammadi H et al. CDK1 in Breast Cancer: Implications for Theranostic Potential. *Anticancer Agents Med Chem*. 2020;20(7):758–767. <https://doi.org/10.2174/1871520620666200203125712>. PMID: 32013835.
70. Brufsky A, Kim SB, Zvirbulis Z, Eniu A, Mebis J, Sohn JH et al. A phase II randomized trial of cobimetinib plus chemotherapy, with or without atezolizumab, as first-line treatment for patients with locally advanced or metastatic triple-negative breast cancer (COLET): primary analysis. *Ann Oncol*. 2021;32(5):652–660. <https://doi.org/10.1016/j.annonc.2021.01.065>. PMID: 33539944.
71. Wawruszak A, Borkiewicz L, Okon E, Kukula-Koch W, Afshan S, Halasa M. Vorinostat (SAHA) and Breast Cancer: An Overview. *Cancers (Basel)*. 2021;13(18):4700. <https://doi.org/10.3390/cancers13184700>. PMID: 34572928.
72. Xu B, Zhang Q, Hu X, Li Q, Sun T, Li W et al. Entinostat, a class I selective histone deacetylase inhibitor, plus exemestane for Chinese patients with hormone receptor-positive advanced breast cancer: A multicenter, randomized, double-blind, placebo-controlled, phase 3 trial. *Acta Pharm Sin B*. 2023;13(5):2250–2258. <https://doi.org/10.1016/j.apsb.2023.02.001>. PMID: 37250148.
73. Johnston SRD, Harbeck N, Hegg R, Toi M, Martin M, Shao ZM et al. Abemaciclib Combined With Endocrine Therapy for the Adjuvant Treatment of HR+, HER2-, Node-Positive, High-Risk, Early Breast Cancer (monarchE). *J Clin Oncol*. 2020;38(34):3987–3998. <https://doi.org/10.1200/JCO.20.02514>. PMID: 32954927.
74. Mayer EL, Dueck AC, Martin M, Rubovszky G, Burstein HJ, Bellet-Ezquerria M et al. Palbociclib with adjuvant endocrine therapy in early breast cancer (PALLAS): interim analysis of a multicentre, open-label, randomised, phase 3 study. *Lancet Oncol*. 2021;22(2):212–222. doi: 10.1016/S1470-2045(20)30642-2. PMID: 33460574.
75. Hortobagyi GN, Stemmer SM, Burris HA, Yap YS, Sonke GS, Paluch-Shimon S et al. Ribociclib as First-Line Therapy for HR-Positive, Advanced Breast Cancer. *N Engl J Med*. 2016;375(18):1738–1748. <https://doi.org/10.1056/NEJMoa1609709>. PMID: 27717303.
76. Arpino G, de la Haba Rodríguez J, Ferrero JM, De Placido S, Osborne CK, Klingbiel D et al. Pertuzumab, Trastuzumab, and an Aromatase Inhibitor for HER2-Positive and Hormone Receptor-Positive Metastatic or Locally Advanced Breast Cancer: PERTAIN Final Analysis. *Clin Cancer Res*. 2023;29(8):1468–1476. <https://doi.org/10.1158/1078-0432.CCR-22-1092>. PMID: 36716289.
77. André F, Ciruelos E, Rubovszky G, Campone M, Loibl S, Rugo HS et al. Alpelisib for PIK3CA-Mutated, Hormone Receptor-Positive Advanced Breast Cancer. *N Engl J Med*. 2019;380(20):1929–1940. <https://doi.org/10.1056/NEJMoa1813904>. PMID: 31091374.
78. Boscolo Bielo L, Trapani D, Nicolò E, Valenza C, Guidi L, Belli C et al. The evolving landscape of metastatic HER2-positive, hormone receptor-positive Breast Cancer. *Cancer Treat Rev*. 2024;128:102761. <https://doi.org/10.1016/j.ctrv.2024.102761>. PMID: 38772169.
79. Wang Z, Che S, Yu Z, PROTAC.: Novel degradable approach for different targets to treat breast cancer. *Eur J Pharm Sci*. 2024;198:106793. <https://doi.org/10.1016/j.ejps.2024.106793>. PMID: 38740076.
80. McCann KE, Hurvitz SA. Advances in the use of PARP inhibitor therapy for breast cancer. *Drugs Context*. 2018;7:212540. <https://doi.org/10.7573/dic.212540>. PMID: 30116283.
81. Pandey P, Khan F, Choi M, Singh SK, Kang HN, Park MN et al. Review deciphering potent therapeutic approaches targeting Notch signaling pathway in breast cancer. *Biomed Pharmacother*. 2023;164:114938. <https://doi.org/10.1016/j.biopha.2023.114938>. PMID: 37267635.
82. Krishna BM, Jana S, Singhal J, Horne D, Awasthi S, Sargia R, Singhal SS. Notch signaling in breast cancer: From pathway analysis to therapy. *Cancer Lett*. 2019;461:123–131. doi: 10.1016/j.canlet.2019.07.012. PMID: 31326555.
83. Marusyk A, Almendro V, Polyak K. Intra-tumour heterogeneity: a looking glass for cancer? *Nat Rev Cancer*. 2012;12(5):323–34. <https://doi.org/10.1038/nrc3261>. PMID: 22513401.
84. Riaz N, Burugu S, Cheng AS, Leung SCY, Gao D, Nielsen TO. Prognostic Significance of CSF-1R Expression in Early Invasive Breast Cancer. *Cancers (Basel)*. 2021;13(22):5769. <https://doi.org/10.3390/cancers13225769>. PMID: 34830923.
85. Baselga J, Cortés J, Kim SB, Im SA, Hegg R, Im YH et al. Pertuzumab plus trastuzumab plus docetaxel for metastatic breast cancer. *N Engl J Med*. 2012;366(2):109–19. <https://doi.org/10.1056/NEJMoa1113216>. PMID: 22149875.
86. Cortés J, Kim SB, Chung WP, Im SA, Park YH, Hegg R et al. Trastuzumab Deruxtecan versus Trastuzumab Emtansine for Breast Cancer. *N Engl J Med*. 2022;386(12):1143–1154. <https://doi.org/10.1056/NEJMoa2115022>. PMID: 35320644.
87. von Minckwitz G, Procter M, de Azambuja E, Zardavas D, Benyunes M, Viale G et al. Adjuvant Pertuzumab and Trastuzumab in Early HER2-Positive Breast Cancer. *N Engl J Med*. 2017;377(2):122–131. <https://doi.org/10.1056/NEJMoa1703643>. PMID: 28581356.
88. Modi S, Jacot W, Yamashita T, Sohn J, Vidal M, Tokunaga E et al. Trastuzumab Deruxtecan in Previously Treated HER2-Low Advanced Breast Cancer. *N Engl J Med*. 2022;387(1):9–20. <https://doi.org/10.1056/NEJMoa2203690>. Epub 2022 Jun 5. PMID: 35665782.
89. Murthy RK, Loi S, Okines A, Piplomata E, Hamilton E, Hurvitz SA et al. Tucatinib, Trastuzumab, and Capecitabine for HER2-Positive Metastatic Breast Cancer. *N Engl J Med*. 2020;382(7):597–609. <https://doi.org/10.1056/NEJMoa1914609>. PMID: 31825569.

90. Li YW, Dai LJ, Wu XR, Zhao S, Xu YZ, Jin X et al. Molecular Characterization and Classification of HER2-Positive Breast Cancer Inform Tailored Therapeutic Strategies. *Cancer Res.* 2024;84(21):3669–3683. <https://doi.org/10.1158/0008-5472.CAN-23-4066>. PMID: 39186675.
91. Subhan MA, Parveen F, Shah H, Yalamarty SSK, Ataide JA, Torchilin VP. Recent Advances with Precision Medicine Treatment for Breast Cancer including Triple-Negative Sub-Type. *Cancers (Basel).* 2023;15(8):2204. <https://doi.org/10.3390/cancers15082204>. PMID: 37190133.
92. Karami Fath M, Azarگونjahromi A, Kiani A, Jalalifar F, Osati P, Akbari Oryani M et al. The role of epigenetic modifications in drug resistance and treatment of breast cancer. *Cell Mol Biol Lett.* 2022;27(1):52. <https://doi.org/10.1186/s11658-022-00344-6>. PMID: 35764927.
93. Wen J, Wang S, Guo R, Liu D. CSF1R inhibitors are emerging immunotherapeutic drugs for cancer treatment. *Eur J Med Chem.* 2023;245(Pt 1):114884. <https://doi.org/10.1016/j.ejmech.2022.114884>. PMID: 36335744.
94. Xu C, Zhao H, Chen H, Yao Q. CXCR4 in breast cancer: oncogenic role and therapeutic targeting. *Drug Des Devel Ther.* 2015;9:4953–64. <https://doi.org/10.2147/DDDT.S84932>. PMID: 26356032.
95. Asleh K, Riaz N, Nielsen TO. Heterogeneity of triple negative breast cancer: Current advances in subtyping and treatment implications. *J Exp Clin Cancer Res.* 2022;41(1):265. <https://doi.org/10.1186/s13046-022-02476-1>. PMID: 36050786.
96. Schmid P, Adams S, Rugo HS, Schneeweiss A, Barrios CH, Iwata H et al. Atezolizumab and Nab-Paclitaxel in Advanced Triple-Negative Breast Cancer. *N Engl J Med.* 2018;379(22):2108–2121. <https://doi.org/10.1056/NEJMoa1809615>. PMID: 30345906.
97. Robson M, Im SA, Senkus E, Xu B, Domchek SM, Masuda N et al. Olaparib for Metastatic Breast Cancer in Patients with a Germline BRCA Mutation. *N Engl J Med.* 2017;377(6):523–533. <https://doi.org/10.1056/NEJMoa1706450>. PMID: 28578601.
98. Litton JK, Rugo HS, Ettl J, Hurvitz SA, Gonçalves A, Lee KH et al. Talazoparib in Patients with Advanced Breast Cancer and a Germline BRCA Mutation. *N Engl J Med.* 2018;379(8):753–763. <https://doi.org/10.1056/NEJMoa1802905>. PMID: 30110579.
99. Jiang YZ, Liu Y, Xiao Y, Hu X, Jiang L, Zuo WJ et al. Molecular subtyping and genomic profiling expand precision medicine in refractory metastatic triple-negative breast cancer: the FUTURE trial. *Cell Res.* 2021;31(2):178–186. <https://doi.org/10.1038/s41422-020-0375-9>. PMID: 32719455.
100. Bardia A, Hurvitz SA, Tolane SM, Lohr D, Punie K, Oliveira M et al. Sacituzumab Govitecan in Metastatic Triple-Negative Breast Cancer. *N Engl J Med.* 2021;384(16):1529–1541. <https://doi.org/10.1056/NEJMoa2028485>. PMID: 33882206.
101. Traina TA, Miller K, Yardley DA, Eakle J, Schwartzberg LS, O'Shaughnessy J et al. Enzalutamide for the Treatment of Androgen Receptor-Expressing Triple-Negative Breast Cancer. *J Clin Oncol.* 2018;36(9):884–890. <https://doi.org/10.1200/JCO.2016.71.3495>. PMID: 29373071.
102. von Minckwitz G, Schneeweiss A, Loibl S, Salat C, Denkert C, Rezai M et al. Neoadjuvant carboplatin in patients with triple-negative and HER2-positive early breast cancer (GeparSixto; GBG 66): a randomised phase 2 trial. *Lancet Oncol.* 2014;15(7):747–56. doi: 10.1016/S1470-2045(14)70160-3. PMID: 24794243.
103. Zhu S, Wu Y, Song B, Yi M, Yan Y, Mei Q et al. Recent advances in targeted strategies for triple-negative breast cancer. *J Hematol Oncol.* 2023;16(1):100. <https://doi.org/10.1186/s13045-023-01497-3>. PMID: 37641116.
104. Tiwari H, Singh S, Sharma S, Gupta P, Verma A, Chattopadhyaya A et al. Deciphering the landscape of triple negative breast cancer from microenvironment dynamics and molecular insights to biomarker analysis and therapeutic modalities. *Med Res Rev.* 2025;45(3):817–841. <https://doi.org/10.1002/med.22090>. PMID: 39445844.
105. Kinnel B, Singh SK, Oprea-Ilie G, Singh R. Targeted Therapy and Mechanisms of Drug Resistance in Breast Cancer. *Cancers (Basel).* 2023;15(4):1320. <https://doi.org/10.3390/cancers15041320>. PMID: 36831661.
106. Moisan A, Madéry M, Boyer T, Domblides C, Blaye C, Larmonier N. Hormone Receptor Signaling and Breast Cancer Resistance to Anti-Tumor Immunity. *Int J Mol Sci.* 2023;24(20):15048. <https://doi.org/10.3390/ijms242015048>. PMID: 37894728.
107. Dong C, Wu J, Chen Y, Nie J, Chen C. Activation of PI3K/AKT/mTOR Pathway Causes Drug Resistance in Breast Cancer. *Front Pharmacol.* 2021;12:628690. <https://doi.org/10.3389/fphar.2021.628690>. PMID: 33790792.
108. Scheidemann ER, Shajahan-Haq AN. Resistance to CDK4/6 Inhibitors in Estrogen Receptor-Positive Breast Cancer. *Int J Mol Sci.* 2021;22(2):12292. <https://doi.org/10.3390/ijms222212292>. PMID: 34830174.
109. Zhou FH, Downton T, Freeland A, Hurwitz J, Caldon CE, Lim E. CDK4/6 inhibitor resistance in estrogen receptor positive breast cancer, a 2023 perspective. *Front Cell Dev Biol.* 2023;11:1148792. <https://doi.org/10.3389/fcell.2023.1148792>. PMID: 37035239.
110. Wang W, Nag SA, Zhang R. Targeting the NFκB signaling pathways for breast cancer prevention and therapy. *Curr Med Chem.* 2015;22(2):264–89. <https://doi.org/10.2174/0929867321666141106124315>. PMID: 25386819.
111. Valadez-Cosmes P, Vázquez-Martínez ER, Cerbón M, Camacho-Arroyo I. Membrane progesterone receptors in reproduction and cancer. *Mol Cell Endocrinol.* 2016;434:166–75. <https://doi.org/10.1016/j.mce.2016.06.027>. PMID: 27368976.
112. Pedroza DA, Subramani R, Lakshmanaswamy R. Classical and Non-Classical Progesterone Signaling in Breast Cancers. *Cancers (Basel).* 2020;12(9):2440. <https://doi.org/10.3390/cancers12092440>. PMID: 32867363.
113. Chen B, Ye P, Chen Y, Liu T, Cha JH, Yan X, Yang WH. Involvement of the Estrogen and Progesterone Axis in Cancer Stemness: Elucidating Molecular Mechanisms and Clinical Significance. *Front Oncol.* 2020;10:1657. <https://doi.org/10.3389/fonc.2020.01657>. PMID: 33014829.
114. Trabert B, Sherman ME, Kannan N, Stanczyk FZ. Progesterone and Breast Cancer. *Endocr Rev.* 2020;41(2):320–44. <https://doi.org/10.1210/edrev/bnz001>. PMID: 31512725.
115. Beato M, Wright RHG, Dily FL. 90 YEARS OF PROGESTERONE: Molecular mechanisms of progesterone receptor action on the breast cancer genome. *J Mol Endocrinol.* 2020;65(1):T65–T79. <https://doi.org/10.1530/JME-19-0266>. PMID: 32485671.
116. Sigl V, Owusu-Boaitey K, Joshi PA, Kavirayani A, Wirnsberger G, Novatchkova M et al. RANKL/RANK control Brca1 mutation-. *Cell Res.* 2016;26(7):761–74. <https://doi.org/10.1038/cr.2016.69>. PMID: 27241552.
117. Lee SR, Lee YH, Jo SL, Heo JH, Kim G, Lee GS et al. Absence of progesterone receptor membrane component 1 reduces migration and metastasis of breast cancer. *Cell Commun Signal.* 2021;19(1):42. <https://doi.org/10.1186/s12964-021-00719-w>. PMID: 33832499.
118. Ferrer-Díaz AI, Sinha G, Petryna A, Gonzalez-Bermejo R, Kenfack Y, Adetayo O et al. Revealing role of epigenetic modifiers and DNA oxidation in cell-autonomous regulation of Cancer stem cells. *Cell Commun Signal.* 2024;22(1):119. <https://doi.org/10.1186/s12964-024-01512-1>. PMID: 38347590.
119. Behrens A, Wurmthaler L, Heindl F, Gass P, Häberle L, Volz B et al. RANK and RANKL Expression in Tumors of Patients with Early Breast Cancer. *Geburtshilfe Frauenheilkd.* 2023;84(1):77–85. <https://doi.org/10.1055/a-2192-2998>. PMID: 38178900.

120. Arteaga CL, Sliwkowski MX, Osborne CK, Perez EA, Puglisi F, Gianni L. Treatment of HER2-positive breast cancer: current status and future perspectives. *Nat Rev Clin Oncol*. 2011;9(1):16–32. <https://doi.org/10.1038/nrclinonc.2011.177>. PMID: 22124364.
121. Garrett JT, Arteaga CL. Resistance to HER2-directed antibodies and tyrosine kinase inhibitors: mechanisms and clinical implications. *Cancer Biol Ther*. 2011;11(9):793–800. <https://doi.org/10.4161/cbt.11.9.15045>. PMID: 21307659.
122. Cortés J, Hurvitz SA, Im SA, Iwata H, Curigliano G et al. Trastuzumab deruxtecan versus trastuzumab emtansine in HER2-positive metastatic breast cancer: long-term survival analysis of the DESTINY-Breast03 trial. *Nat Med*. 2024;30(8):2208–2215. <https://doi.org/10.1038/s41591-024-03021-7>. PMID: 38825627.
123. Raghav KPS, Moasser MM. Molecular Pathways and Mechanisms of HER2 in Cancer Therapy. *Clin Cancer Res*. 2023;29(13):2351–2361. <https://doi.org/10.1158/1078-0432.CCR-22-0283>. PMID: 36574481.
124. Pourjamal N, Yazdi N, Halme A, Joncour VL, Laakkonen P, Saharinen P et al. Comparison of trastuzumab emtansine, trastuzumab deruxtecan, and disitamab vedotin in a multiresistant HER2-positive breast cancer lung metastasis model. *Clin Exp Metastasis*. 2024;41(2):91–102. <https://doi.org/10.1007/s10585-024-10278-2>. PMID: 38367127.
125. Rexer BN, Arteaga CL. Intrinsic and acquired resistance to HER2-targeted therapies in HER2 gene-amplified breast cancer: mechanisms and clinical implications. *Crit Rev Oncog*. 2012;17(1):1–16. <https://doi.org/10.1615/critrevoncog.v17.i1.20>. PMID: 22471661.
126. Hong R, Xu B. Breast cancer: an up-to-date review and future perspectives. *Cancer Commun (Lond)*. 2022;42(10):913–936. <https://doi.org/10.1002/cac2.12358>. PMID: 36074908.
127. Bose R, Kavuri SM, Searleman AC, Shen W, Shen D, Koboldt DC et al. Activating HER2 mutations in HER2 gene amplification negative breast cancer. *Cancer Discov*. 2013;3(2):224–37. <https://doi.org/10.1158/2159-8290.CD-12-0349>. PMID: 23220880.
128. Dvir K, Giordano S, Leone JP. Immunotherapy in Breast Cancer. *Int J Mol Sci*. 2024;25(14):7517. <https://doi.org/10.3390/ijms25147517>. PMID: 39062758.
129. Kaufman B, Mackey JR, Clemens MR, Bapsy PP, Vaid A, Wardley A et al. Trastuzumab plus anastrozole versus anastrozole alone for the treatment of postmenopausal women with human epidermal growth factor receptor 2-positive, hormone receptor-positive metastatic breast cancer: results from the randomized phase III TAnDEM study. *J Clin Oncol*. 2009;27(33):5529–37. <https://doi.org/10.1200/JCO.2008.20.6847>. PMID: 19786670.
130. Rugo HS, Im SA, Cardoso F, Cortes J, Curigliano G, Musolino A, et al. Margetuximab versus trastuzumab in patients with previously treated HER2-Positive advanced breast cancer (SOPHIA): final overall survival results from a randomized phase 3 trial. *J Clin Oncol*. 2023;41(2):198–205. PMID: 36332179; PMCID: PMC9839304.
131. Hao C, Wei Y, Meng W, Zhang J, Yang X. PI3K/AKT/mTOR inhibitors for hormone receptor-positive advanced breast cancer. *Cancer Treat Rev*. 2025;132:102861. <https://doi.org/10.1016/j.ctrv.2024.102861>. PMID: 39662202.
132. Scaltriti M, Rojo F, Ocaña A, Anido J, Guzman M, Cortes J et al. Expression of p95HER2, a truncated form of the HER2 receptor, and response to anti-HER2 therapies in breast cancer. *J Natl Cancer Inst*. 2007;99(8):628–38. <https://doi.org/10.1093/jnci/djk134>. PMID: 17440164.
133. Huang P, Zhou X, Zheng M, Yu Y, Jin G, Zhang S. Regulatory T cells are associated with the tumor immune microenvironment and immunotherapy response in triple-negative breast cancer. *Front Immunol*. 2023;14:1263537. <https://doi.org/10.3389/fimmu.2023.1263537>. PMID: 37767092.
134. Chen X, Feng L, Huang Y, Wu Y, Xie N. Mechanisms and Strategies to Overcome PD-1/PD-L1 Blockade Resistance in Triple-Negative Breast Cancer. *Cancers (Basel)*. 2022;15(1):104. <https://doi.org/10.3390/cancers15010104>. PMID: 36612100.
135. Arpino G, Wiechmann L, Osborne CK, Schiff R. Crosstalk between the estrogen receptor and the HER tyrosine kinase receptor family: molecular mechanism and clinical implications for endocrine therapy resistance. *Endocr Rev*. 2008;29(2):217–33. <https://doi.org/10.1210/er.2006-0045>. PMID: 18216219.
136. Pegram M, Jackisch C, Johnston SRD. Estrogen/HER2 receptor crosstalk in breast cancer: combination therapies to improve outcomes for patients with hormone receptor-positive/HER2-positive breast cancer. *NPJ Breast Cancer*. 2023;9(1):45. <https://doi.org/10.1038/s41523-023-00533-2>. PMID: 37258523.
137. Bowen J, Braga S, Zotto VD, Finnie J, DiPrimeo D, Cooke B et al. Preclinical and clinical evaluation through serial colonoscopic evaluation of neratinib-induced diarrhea in HER2-positive breast cancer-A pilot study. *Physiol Rep*. 2024;12(16):e70008. <https://doi.org/10.14814/phy2.70008>. PMID: 39187401.
138. van Ramshorst MS, van Werkhoven ED, Mandjes IA, Kemper I, Dezentjé VO et al. Neoadjuvant chemotherapy with or without anthracyclines in the presence of dual HER2 blockade for HER2-positive breast cancer (TRAIN-2): a multicentre, open-label, randomised, phase 3 trial. *Lancet Oncol*. 2018;19(12):1630–1640. [https://doi.org/10.1016/S1470-2045\(18\)30570-9](https://doi.org/10.1016/S1470-2045(18)30570-9). PMID: 30413379.
139. Finn RS, Martin M, Rugo HS, Jones S, Im SA, Gelmon K et al. Palbociclib and Letrozole in Advanced Breast Cancer. *N Engl J Med*. 2016;375(20):1925–1936. <https://doi.org/10.1056/NEJMoa1607303>. PMID: 27959613.
140. Nunnery SE, Mayer IA. Targeting the PI3K/AKT/mTOR Pathway in Hormone-Positive Breast Cancer. *Drugs*. 2020;80(16):1685–1697. <https://doi.org/10.1007/s40265-020-01394-w>. PMID: 32894420.
141. Manore SG, Doherty DL, Wong GL, Lo HW. IL-6/JAK/STAT3 Signaling in Breast Cancer Metastasis: Biology and Treatment. *Front Oncol*. 2022;12:866014. <https://doi.org/10.3389/fonc.2022.866014>. PMID: 35371975.
142. Guerrero-Zotano A, Mayer IA, Arteaga CL. PI3K/AKT/mTOR: role in breast cancer progression, drug resistance, and treatment. *Cancer Metastasis Rev*. 2016;35(4):515–524. <https://doi.org/10.1007/s10555-016-9637-x>. PMID: 27896521.
143. Miricescu D, Totan A, Stanescu-Spinu II, Badoiu SC, Stefani C, Greabu M. PI3K/AKT/mTOR Signaling Pathway in Breast Cancer: From Molecular Landscape to Clinical Aspects. *Int J Mol Sci*. 2020;22(1):173. <https://doi.org/10.3390/ijms22010173>. PMID: 33375317.
144. Zhao Y, Murciano-Goroff YR, Xue JY, Ang A, Lucas J, Mai TT et al. Diverse alterations associated with resistance to KRAS(G12C) inhibition. *Nature*. 2021;599(7886):679–683. <https://doi.org/10.1038/s41586-021-04065-2>. PMID: 34759319.
145. Cheng J, Song B, Wei C, Zhang L, Liu X, Yang L et al. Exploring breast cancer associated-gene panel for next-generation sequencing and identifying new, pathogenic variants in breast cancer from western China. *J Cancer*. 2025;16(4):1281–1295. <https://doi.org/10.7150/jca.101911>. PMID: 39895796.
146. Zhao ZM, Yost SE, Hutchinson KE, Li SM, Yuan YC, Noorbakhsh J et al. CCNE1 amplification is associated with poor prognosis in patients with triple negative breast cancer. *BMC Cancer*. 2019;19(1):96. <https://doi.org/10.1186/s12885-019-5290-4>. PMID: 30665374.

147. Yan D, Li G, Yuan Y, Li H, Cao H, Dai Y et al. SOCS3 inhibiting JAK-STAT pathway enhances oncolytic adenovirus efficacy by potentiating viral replication and T-cell activation. *Cancer Gene Ther.* 2024;31(3):397–409. <https://doi.org/10.1038/s41417-023-00710-2>. PMID: 38102464.
148. McCubrey JA, Steelman LS, Abrams SL, Lee JT, Chang F, Bertrand FE et al. Roles of the RAF/MEK/ERK and PI3K/PTEN/AKT pathways in malignant transformation and drug resistance. *Adv Enzyme Regul.* 2006;46:249–79. <https://doi.org/10.1016/j.advenzreg.2006.01.004>. PMID: 16854453.
149. Qiu D, Yan X, Xiao X, Zhang G, Wang Y, Cao J et al. To explore immune synergistic function of Quercetin in inhibiting breast cancer cells. *Cancer Cell Int.* 2021;21(1):632. <https://doi.org/10.1186/s12935-021-02345-5>. PMID: 34838003.
150. Alsherbiny MA, Bhuyan DJ, Low MN, Chang D, Li CG. Synergistic Interactions of Cannabidiol with Chemotherapeutic Drugs in MCF7 Cells: Mode of Interaction and Proteomics Analysis of Mechanisms. *Int J Mol Sci.* 2021;22(18):10103. <https://doi.org/10.3390/ijms221810103>. PMID: 34576262.
151. Kaur H, Ghosh S, Kumar P, Basu B, Nagpal K. Ellagic acid-loaded, tween 80-coated, chitosan nanoparticles as a promising therapeutic approach against breast cancer: In-vitro and in-vivo study. *Life Sci.* 2021;284:119927. <https://doi.org/10.1016/j.lfs.2021.119927>. PMID: 34492262.
152. Tavakoli M, Maghsoudian S, Rezaei-Aderiani A, Hajiramezani M, Fatahi Y, Amani M et al. Synergistic effects of paclitaxel and platelet-superparamagnetic iron oxide nanoparticles for targeted chemo-hyperthermia therapy against breast cancer. *Colloids Surf B Biointerfaces.* 2025;251:114584. <https://doi.org/10.1016/j.colsurfb.2025.114584>. PMID: 39999697.
153. Jain A, Barge A, Parris CN. Combination strategies with PARP inhibitors in BRCA-mutated triple-negative breast cancer: overcoming resistance mechanisms. *Oncogene.* 2025;44(4):193–207. <https://doi.org/10.1038/s41388-024-03227-6>. PMID: 39572842.
154. Sun Y, Fang K, Hu X, Yang J, Jiang Z, Feng L et al. NIR-light-controlled G-quadruplex hydrogel for synergistically enhancing photodynamic therapy via sustained delivery of metformin and catalase-like activity in breast cancer. *Mater Today Bio.* 2022;16:100375. <https://doi.org/10.1016/j.mtbio.2022.100375>. PMID: 35983175.
155. Yi GY, Kim MJ, Kim HI, Park J, Baek SH. Hyperthermia Treatment as a Promising Anti-Cancer Strategy: Therapeutic Targets, Perspective Mechanisms and Synergistic Combinations in Experimental Approaches. *Antioxidants (Basel).* 2022;11(4):625. <https://doi.org/10.3390/antiox11040625>. PMID: 35453310.
156. Duarte D, Cardoso A, Vale N. Synergistic Growth Inhibition of HT-29 Colon and MCF-7 Breast Cancer Cells with Simultaneous and Sequential Combinations of Antineoplastics and CNS Drugs. *Int J Mol Sci.* 2021;22(14):7408. <https://doi.org/10.3390/ijms22147408>. PMID: 34299028.
157. Xu C, Dong J, Shi X, Rui J, Chen M, Lu W et al. Engineered microalgae for photo-sonodynamic synergistic therapy in breast cancer treatment. *Acta Biomater.* 2025;193:531–544. doi: 10.1016/j.actbio.2024.12.047. PMID: 39709158.
158. Haque S, Cook K, Sahay G, Sun C. RNA-Based Therapeutics: Current Developments in Targeted Molecular Therapy of Triple-Negative Breast Cancer. *Pharmaceutics.* 2021;13(10):1694. <https://doi.org/10.3390/pharmaceutics13101694>. PMID: 34683988.
159. Wilson Z, Odedra R, Wallez Y, Wijnhoven PWG, Hughes AM, Gerrard J et al. ATR Inhibitor AZD6738 (Ceralasertib) Exerts Anti-tumor Activity as a Monotherapy and in Combination with Chemotherapy and the PARP Inhibitor Olaparib. *Cancer Res.* 2022;82(6):1140–1152. <https://doi.org/10.1158/0008-5472.CAN-21-2997>. PMID: 35078817.
160. Brett JO, Spring LM, Bardia A, Wander SA. ESR1 mutation as an emerging clinical biomarker in metastatic hormone receptor-positive breast cancer. *Breast Cancer Res.* 2021;23(1):85. <https://doi.org/10.1186/s13058-021-01462-3>. PMID: 34392831.
161. Rasha F, Sharma M, Pruitt K. Mechanisms of endocrine therapy resistance in breast cancer. *Mol Cell Endocrinol.* 2021;532:111322. <https://doi.org/10.1016/j.mce.2021.111322>. PMID: 34000350.
162. Tufail M, Cui J, Wu C. Breast cancer: molecular mechanisms of underlying resistance and therapeutic approaches. *Am J Cancer Res.* 2022;12(7):2920–49. PMID: 35968356.
163. Chen YF, Xu YY, Shao ZM, Yu KD. Resistance to antibody-drug conjugates in breast cancer: mechanisms and solutions. *Cancer Commun (Lond).* 2023;43(3):297–337. <https://doi.org/10.1002/cac2.12387>. Epub 2022 Nov 10. PMID: 36357174.
164. Kurimchak AM, Herrera-Montáñez C, Montserrat-Sangrà S, Araiza-Olivera D, Hu J, Neumann-Domer R et al. The drug efflux pump MDR1 promotes intrinsic and acquired resistance to PROTACs in cancer cells. *Sci Signal.* 2022;15(749):eabn2707. <https://doi.org/10.1126/scisignal.abn2707>. PMID: 36041010.
165. Turner N, Huang-Bartlett C, Kalinsky K, Cristofanilli M, Bianchini G, Chia S, et al. Design of SERENA-6, a phase III switching trial of camizestrant in ESR1-mutant breast cancer during first-line treatment. *Future Oncol.* 2023;19(8):559–73. <https://doi.org/10.2217/fon-2022-1196>. PMID: 36309080.
166. Zhang Z, Richmond A, Yan C. Immunomodulatory Properties of PI3K/AKT/mTOR and MAPK/MEK/ERK Inhibition Augment Response to Immune Checkpoint Blockade in Melanoma and Triple-Negative Breast Cancer. *Int J Mol Sci.* 2022;23(13):7353. <https://doi.org/10.3390/ijms23137353>. PMID: 35806358.
167. Li G, Lin SS, Yu ZL, Wu XH, Liu JW, Tu GH et al. A PARP1 PROTAC as a novel strategy against PARP inhibitor resistance via promotion of ferroptosis in p53-positive breast cancer. *Biochem Pharmacol.* 2022;206:115329. <https://doi.org/10.1016/j.bcp.2022.115329>. PMID: 36309080.
168. Roarty K, Echeverria GV. Laboratory Models for Investigating Breast Cancer Therapy Resistance and Metastasis. *Front Oncol.* 2021;11:645698. <https://doi.org/10.3389/fonc.2021.645698>. PMID: 33777805.
169. Adam-Artigues A, Arenas EJ, Martínez-Sabadell A, Brasó-Maristany F, Cervera R, Tormo E, et al. Targeting HER2-AXL heterodimerization to overcome resistance to HER2 Blockade in breast cancer. *Sci Adv.* 2022;8(20):eabk2746. <https://doi.org/10.1126/sciadv.abk2746>. PMID: 36309080.
170. Palafox M, Monserrat L, Bellet M, Villacampa G, Gonzalez-Perez A, Oliveira M et al. High p16 expression and heterozygous RB1 loss are biomarkers for CDK4/6 inhibitor resistance in ER+ breast cancer. *Nat Commun.* 2022;13(1):5258. <https://doi.org/10.1038/s41467-022-32828-6>. PMID: 36071033.
171. Stevens LE, Peluffo G, Qiu X, Temko D, Fassl A, Li Z et al. JAK-STAT Signaling in Inflammatory Breast Cancer Enables Chemotherapy-Resistant Cell States. *Cancer Res.* 2023;83(2):264–284. <https://doi.org/10.1158/0008-5472.CAN-22-0423>. PMID: 36409824.
172. Zou Y, Zheng S, Xie X, Ye F, Hu X, Tian Z et al. N6-methyladenosine regulated FGFR4 attenuates ferroptotic cell death in recalcitrant HER2-positive breast cancer. *Nat Commun.* 2022;13(1):2672. <https://doi.org/10.1038/s41467-022-30217-7>. PMID: 35562334.

173. Liang Y, Wang Y, Zhang Y, Ye F, Luo D, Li Y et al. HSPB1 facilitates chemoresistance through inhibiting ferroptotic cancer cell death and regulating NF- κ B signaling pathway in breast cancer. *Cell Death Dis.* 2023;14(7):434. <https://doi.org/10.1038/s41419-023-05972-0>. PMID: 37454220.
174. Wang B, Zhang W, Zhang G, Kwong L, Lu H, Tan J et al. Targeting mTOR signaling overcomes acquired resistance to combined BRAF and MEK inhibition in BRAF-mutant melanoma. *Oncogene.* 2021;40(37):5590–5599. <https://doi.org/10.1038/s41388-021-01911-5>. PMID: 34304249.
175. Dang F, Nie L, Zhou J, Shimizu K, Chu C, Wu Z et al. Inhibition of CK1 ϵ potentiates the therapeutic efficacy of CDK4/6 inhibitor in breast cancer. *Nat Commun.* 2021;12(1):5386. <https://doi.org/10.1038/s41467-021-25700-6>. PMID: 34508104.
176. Xiao M, He J, Yin L, Chen X, Zu X, Shen Y. Tumor-Associated Macrophages: Critical Players in Drug Resistance of Breast Cancer. *Front Immunol.* 2021;12:799428. <https://doi.org/10.3389/fimmu.2021.799428>. PMID: 34992609.
177. Deepak KGK, Vempati R, Nagaraju GP, Dasari VR, Rao SN et al. DN α , Tumor microenvironment: Challenges and opportunities in targeting metastasis of triple negative breast cancer. *Pharmacol Res.* 2020;153:104683. <https://doi.org/10.1016/j.phrs.2020.104683>. PMID: 32050092.
178. Salvadori G, Zanardi F, Iannelli F, Lobefaro R, Vernieri C, Longo VD. Fasting-mimicking diet blocks triple-negative breast cancer and cancer stem cell escape. *Cell Metab.* 2021;33(1):2247–2259.e6. <https://doi.org/10.1016/j.cmet.2021.10.008>. PMID: 34731655.
179. Raheem F, Karikalan SA, Batalini F, El Masry A, Mina L. Metastatic ER+ Breast Cancer: Mechanisms of Resistance and Future Therapeutic Approaches. *Int J Mol Sci.* 2023;24(22):16198. <https://doi.org/10.3390/ijms242216198>. PMID: 38003387.
180. Lei P, Wang W, Sheldon M, Sun Y, Yao F, Ma L. Role of Glucose Metabolic Reprogramming in Breast Cancer Progression and Drug Resistance. *Cancers (Basel).* 2023;15(13):3390. <https://doi.org/10.3390/cancers15133390>. PMID: 37444501.
181. Cai Z, Shi Q, Li Y, Jin L, Li S, Wong LL et al. LncRNA EILA promotes CDK4/6 inhibitor resistance in breast cancer by stabilizing cyclin E1 protein. *Sci Adv.* 2023;9(40):eadi3821. <https://doi.org/10.1126/sciadv.adi3821>. PMID: 37801505.
182. Manousakis E, Miralles CM, Esquerda MG, Wright RHG. CDKN1A/p21 in Breast Cancer: Part of the Problem, or Part of the Solution? *Int J Mol Sci.* 2023;24(24):17488. <https://doi.org/10.3390/ijms242417488>. PMID: 38139316.
183. Chen H, Jia B, Zhang Q, Zhang Y. Meclofenamic Acid Restores Gefitinib Sensitivity by Downregulating Breast Cancer Resistance Protein and Multidrug Resistance Protein 7 via FTO/m6A-Demethylation/c-Myc in Non-Small Cell Lung Cancer. *Front Oncol.* 2022;12:870636. <https://doi.org/10.3389/fonc.2022.870636>. PMID: 35530301.
184. Siatis KE, Giannopoulou E, Manou D, Sarantis P, Karamouzis MV, Raftopoulou S et al. Resistance to hormone therapy in breast cancer cells promotes autophagy and EGFR signaling pathway. *Am J Physiol Cell Physiol.* 2023;325(3):C708–C720. <https://doi.org/10.1152/ajpcell.001992023>. PMID: 37575061.
185. Smith AE, Ferraro E, Safonov A, Morales CB, Lahuerta EJA, Li Q et al. HER2+ breast cancers evade anti-HER2 therapy via a switch in driver pathway. *Nat Commun.* 2021;12(1):6667. <https://doi.org/10.1038/s41467-021-27093-y>. PMID: 34795269.
186. Janiszewska M, Stein S, Metzger Filho O, Eng J, Kingston NL, Harper NW et al. The impact of tumor epithelial and microenvironmental heterogeneity on treatment responses in HER2+ breast cancer. *JCI Insight.* 2021;6(11):e147617. <https://doi.org/10.1172/jci.insight.147617>. PMID: 33886505.
187. Sahoo S, Mishra A, Kaur H, Hari K, Muralidharan S, Mandal S et al. A mechanistic model captures the emergence and implications of non-genetic heterogeneity and reversible drug resistance in ER+ breast cancer cells. *NAR Cancer.* 2021;3(3):zcab027. <https://doi.org/10.1093/narcan/zcab027>. PMID: 34316714.
188. Chen B, Dragomir MP, Yang C, Li Q, Horst D, Calin GA. Targeting non-coding RNAs to overcome cancer therapy resistance. *Signal Transduct Target Ther.* 2022;7(1):121. <https://doi.org/10.1038/s41392-022-00975-3>. PMID: 35418578.
189. Obidiro O, Battogtokh G, Akala EO. Triple Negative Breast Cancer Treatment Options and Limitations: Future Outlook. *Pharmaceutics.* 2023;15(7):1796. <https://doi.org/10.3390/pharmaceutics15071796>. PMID: 37513983.
190. Zhao Z, Ma X, Cai Z. The potential role of CD8+ cytotoxic T lymphocytes and one branch connected with tissue-resident memory in non-luminal breast cancer. *PeerJ.* 2024;12:e17667. <https://doi.org/10.7717/peerj.17667>. PMID: 39006029.
191. Jin M, Fang J, Peng J, Wang X, Xing P, Jia K et al. PD-1/PD-L1 immune checkpoint blockade in breast cancer: research insights and sensitization strategies. *Mol Cancer.* 2024;23(1):266. <https://doi.org/10.1186/s12943-024-02176-8>. PMID: 39614285.
192. Stavrou M, Constantinidou A. Tumor associated macrophages in breast cancer progression: implications and clinical relevance. *Front Immunol.* 2024;15:1441820. <https://doi.org/10.3389/fimmu.2024.1441820>. PMID: 39044824.
193. Huang X, Cao J, Zu X. Tumor-associated macrophages: An important player in breast cancer progression. *Thorac Cancer.* 2022;13(3):269–276. <https://doi.org/10.1111/1759-7714.14268>. PMID: 34914196.
194. Sahai E, Astsaturov I, Cukierman E, DeNardo DG, Egeblad M, Evans RM et al. A framework for advancing our understanding of cancer-associated fibroblasts. *Nat Rev Cancer.* 2020;20(3):174–186. <https://doi.org/10.1038/s41568-019-0238-1>. PMID: 31980749.
195. Banerjee S. The combination of VEGF inhibitors and anti-oestrogen therapies in breast cancer. *Steroids.* 2011;76(8):807–11. <https://doi.org/10.1016/j.steroids.2011.02.036>. PMID: 21439993.
196. Baylin SB, Jones PA. Epigenetic Determinants of Cancer. *Cold Spring Harb Perspect Biol.* 2016;8(9):a019505. <https://doi.org/10.1101/cshperspect.a019505>. PMID: 27194046.
197. Jones PA, Issa JP, Baylin S. Targeting the cancer epigenome for therapy. *Nat Rev Genet.* 2016;17(10):630–41. <https://doi.org/10.1038/nrg.2016.93>. PMID: 27629931.
198. Ediriweera MK, Tennekoon KH, Samarakoon SR. Emerging role of histone deacetylase inhibitors as anti-breast-cancer agents. *Drug Discov Today.* 2019;24(3):685–702. <https://doi.org/10.1016/j.drudis.2019.02.003>. PMID: 30776482.
199. Li Y, Seto E. HDACs and HDAC Inhibitors in Cancer Development and Therapy. *Cold Spring Harb Perspect Med.* 2016;6(10):a026831. <https://doi.org/10.1101/cshperspect.a026831>. PMID: 27599530.
200. Topper MJ, Vaz M, Chiappinelli KB, DeStefano Shields CE, Niknafs N et al. Epigenetic Therapy Ties MYC Depletion to Reversing Immune Evasion and Treating Lung Cancer. *Cell.* 2017;171(6):1284–1300.e21. <https://doi.org/10.1016/j.cell.2017.10.022>. PMID: 29195073.
201. Wang Y, Bu N, Luan XF, Song QQ, Ma BF, Hao W et al. Harnessing the potential of long non-coding RNAs in breast cancer: from etiology to treatment resistance and clinical applications. *Front Oncol.* 2024;14:1337579. <https://doi.org/10.3389/fonc.2024.1337579>. PMID: 38505593.
202. Liu S, Zhang X, Wang W, Li X, Sun X, Zhao Y et al. Metabolic reprogramming and therapeutic resistance in primary and metastatic breast cancer. *Mol Cancer.* 2024;23(1):261. <https://doi.org/10.1186/s12943-024-02165-x>. PMID: 39574178.

203. Gandhi N, Das GM. Metabolic Reprogramming in Breast Cancer and Its Therapeutic Implications. *Cells*. 2019;8(2):89. <https://doi.org/10.3390/cells8020089>. PMID: 30691108.
204. Liu Y, Zong X, Altea-Manzano P, Fu J. Amino acid metabolism in breast cancer: pathogenic drivers and therapeutic opportunities. *Protein Cell*. 2025;16(7):506–531. <https://doi.org/10.1093/procel/pwaf011>. PMID: 39973065.
205. Xiao Q, Xia M, Tang W, Zhao H, Chen Y, Zhong J. The lipid metabolism remodeling: A hurdle in breast cancer therapy. *Cancer Lett*. 2024;582:216512. <https://doi.org/10.1016/j.canlet.2023.216512>. PMID: 38036043.
206. Garg P, Singhal G, Horne D, Salgia R, Singhal SS. Metabolic reprogramming in breast cancer: Pathways driving progression, drug resistance, and emerging therapeutics. *Biochim Biophys Acta Rev Cancer*. 2025;1880(5):189396. <https://doi.org/10.1016/j.bbcan.2025.189396>. PMID: 40701505.
207. Kundu M, Butti R, Panda VK, Malhotra D, Das S, Mitra T et al. Modulation of the tumor microenvironment and mechanism of immunotherapy-based drug resistance in breast cancer. *Mol Cancer*. 2024;23(1):92. <https://doi.org/10.1186/s12943-024-01990-4>. PMID: 38715072.
208. Yan W, Wu X, Zhou W, Fong MY, Cao M, Liu J et al. Cancer-cell-secreted exosomal miR-105 promotes tumour growth through the MYC-dependent metabolic reprogramming of stromal cells. *Nat Cell Biol*. 2018;20(5):597–609. <https://doi.org/10.1038/s41556-018-0083-6>. PMID: 29662176.
209. Mebane RH, Noel T, Ing N, Gouin KH 3rd, Shah A, Zitser D et al. Spatial transcriptomic analysis of immune checkpoint blockade response in triple negative breast cancers with tertiary lymphoid structures. *iScience*. 2025;28(7):112808. <https://doi.org/10.1016/j.isci.2025.112808>. PMID: 40599321.
210. Mao X, Zhou D, Lin K, Zhang B, Gao J, Ling F, et al. Single-cell and Spatial transcriptome analyses revealed cell heterogeneity and immune environment alternations in metastatic axillary lymph nodes in breast cancer. *Cancer Immunol Immunother*. 2023;72(3):679–95. <https://doi.org/10.1007/s00262-022-03278-2>. Epub 2022 Aug 30. PMID: 36040519.
211. Andersson A, Larsson L, Stenbeck L, Salmén F, Ehinger A, Wu SZ et al. Spatial deconvolution of HER2-positive breast cancer delineates tumor-associated cell type interactions. *Nat Commun*. 2021;12(1):6012. <https://doi.org/10.1038/s41467-021-26271-2>. PMID: 34650042.
212. Chu X, Tian Y, Lv C. Decoding the spatiotemporal heterogeneity of tumor-associated macrophages. *Mol Cancer*. 2024;23(1):150. <https://doi.org/10.1186/s12943-024-02064-1>. PMID: 39068459.
213. Nasir I, McGuinness C, Poh AR, Ernst M, Darcy PK, Britt KL. Tumor macrophage functional heterogeneity can inform the development of novel cancer therapies. *Trends Immunol*. 2023;44(12):971–985. <https://doi.org/10.1016/j.it.2023.10.007>. PMID: 37995659.
214. Wang L, Guo W, Guo Z, Yu J, Tan J, Simons DL et al. PD-L1-expressing tumor-associated macrophages are immunostimulatory and associate with good clinical outcome in human breast cancer. *Cell Rep Med*. 2024;5(2):101420. <https://doi.org/10.1016/j.xcrm.2024.101420>. PMID: 38382468.
215. Verona F, Di Bella S, Schirano R, Manfredi C, Angeloro F, Bozzari G et al. Cancer stem cells and tumor-associated macrophages as mates in tumor progression: mechanisms of crosstalk and advanced bioinformatic tools to dissect their phenotypes and interaction. *Front Immunol*. 2025;16:1529847. <https://doi.org/10.3389/fimmu.2025.1529847>. PMID: 39981232.
216. Gao Y, Li B, Jin Y, Cheng J, Tian W, Ying L et al. Spatial multi-omics profiling of breast cancer oligo-recurrent lung metastasis. *Oncogene*. 2025;44(27):2268–2282. <https://doi.org/10.1038/s41388-025-03388-y>. PMID: 40234722.
217. Zhang Y, Gong S, Liu X. Spatial transcriptomics: a new frontier in accurate localization of breast cancer diagnosis and treatment. *Front Immunol*. 2024;15:1483595. <https://doi.org/10.3389/fimmu.2024.1483595>. PMID: 39439806.
218. Sun Y, Pan Z, Wang Z, et al. Single-cell and Spatial transcriptome analysis reveals the cellular heterogeneity of liver metastatic colorectal cancer. *J Transl Med*. 2024;22(1):894. <https://doi.org/10.1186/s12967-024-05706-6>.
219. Ahn JS, Shin S, Yang SA, Park EK, Kim KH, Cho SI et al. Artificial Intelligence in Breast Cancer Diagnosis and Personalized Medicine. *J Breast Cancer*. 2023;26(5):405–435. <https://doi.org/10.4048/jbc.2023.26.e45>. PMID: 37926067.
220. Alum EU. AI-driven biomarker discovery: enhancing precision in cancer diagnosis and prognosis. *Discov Oncol*. 2025;16(1):313. <https://doi.org/10.1007/s12672-025-02064-7>. PMID: 40082367; PMCID: PMC11906928.
221. Bhattacharya S, Saleem SM, Singh A, Singh S, Tripathi S. Empowering precision medicine: regenerative AI in breast cancer. *Front Oncol*. 2024;14:1465720. <https://doi.org/10.3389/fonc.2024.1465720>. PMID: 39372870.
222. Srivastava R. Advancing precision oncology with AI-powered genomic analysis. *Front Pharmacol*. 2025;16:1591696. <https://doi.org/10.3389/fphar.2025.1591696>. PMID: 40371349.
223. Yan S, Li J, Wu W. Artificial intelligence in breast cancer: application and future perspectives. *J Cancer Res Clin Oncol*. 2023;149(17):16179–16190. <https://doi.org/10.1007/s00432-023-05337-2>. PMID: 37656245.
224. Sobhani N, D'Angelo A, Pittacolo M, Mondani G, Generali D. Future AI Will Most Likely Predict Antibody-Drug Conjugate Response in Oncology: A Review and Expert Opinion. *Cancers (Basel)*. 2024;16(17):3089. <https://doi.org/10.3390/cancers16173089>. PMID: 39272947.
225. Mkhize-Kwitshana Z, Naidoo P, Duma Z, Saini KS, Dlamini Z. Revolutionizing breast cancer treatment: harnessing the power of artificial intelligence in overcoming drug resistance. In: *Overcoming Breast Cancer Therapy Resistance: From Mechanisms to Precision and AI-Powered Approaches*. 2024. p 235–57. https://doi.org/10.1007/978-3-031-52860-6_10
226. Shirzad M, Shaban M, Mohammadzadeh V, Rahdar A, Fathi-karkan S, Hoseini ZS, et al. Artificial intelligence-assisted design of nanomedicines for breast cancer diagnosis and therapy: advances, challenges, and future directions. *BioNanoScience*. 2025;15(3):1–33. <https://doi.org/10.1007/s12668-025-01980-w>.

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